T.C.
DOKUZ EYLUL UNIVERSITY
ENGINEERING FACULTY
ELECTRICAL&ELECTRONICS ENGINEERING
DEPARTMENT

MODEL BASED DESIGN OF IMAGE ENHANCEMENT TECHNIQUES FOR FPGA IMPLEMENTATION

Final Year Project

by

Abdullah SÖNMEZ

Advisor

Prof. Dr. Uğur ÇAM

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IZMIR

THESIS EVALUATION FORM

We certify that we have read this thesis and that in our opinion it is fully adequate, in scope and qualify as an undergraduate thesis, based on the result of the oral examination taken place on .../.../2025.

.....

Prof. Dr. Uğur ÇAM

(Advisor)

.....	.....
Prof. Dr. Özge CİHANBEĞENDİ	Prof. Dr. Taner Abdullah OĞUZER
(Committee Member)	(Committee Member)

.....

Prof. Dr. Yavuz ŞENOL

(Chairman)

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I would also like to thanks my family who are a source of motivation for me.

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ABSTRACT

Image processing plays an important role in many application areas and is a key component of image enhancement. In this study, the application of the histogram equalization technique on the MATLAB-Simulink platform for image enhancement was examined. Histogram equalization increases the contrast of images, making them more distinct and improving visual quality. The MATLAB-Simulink environment allows effective modelling, simulation and analysis of this technique.

In this study, firstly, the histogram equalization algorithm was developed in MATLAB and then a model was created in the Simulink environment. MATLAB and Simulink provide an ideal platform to quickly and effectively prototype and test image processing algorithms. The histogram equalization model was tested on different types of images and its performance was evaluated.

Next, the design was made for real-time application on Field Programmable Gate Arrays (FPGA). FPGA-based applications are ideal for real-time image processing due to their high speed and parallel processing capabilities. The histogram equalization model was implemented on the FPGA using Hardware Description Language (HDL) and its performance was evaluated on the FPGA platform.

This work concludes by outlining a methodology for real-time implementation of the histogram equalization technique for picture enhancement on an FPGA and modeling it in the MATLAB-Simulink environment. A model-based design technique was used to perform histogram equalization; Vivado software was then used to synthesize and simulate the SystemVerilog hardware description code that was created from this model. This approach is a useful tool for enhancing image quality and can be applied to image processing applications.

ÖZET

Görüntü işleme, birçok uygulama alanında önemli bir rol oynamaktadır ve görüntü iyileştirmenin temel bir bileşenidir. Bu çalışmada, görüntü iyileştirme için histogram eşitleme tekniğinin MATLAB-Simulink platformu üzerindeki uygulaması incelenmiştir. Histogram eşitleme, görüntülerin kontrastını artırarak daha belirgin hale getirir ve görsel kaliteyi iyileştirir. MATLAB-Simulink ortamı, bu tekniğin etkili bir şekilde modellemesi, simülasyonu ve analizine olanak tanır.

Bu çalışmada, öncelikle MATLAB'de histogram eşitleme algoritması geliştirilmiş ve ardından Simulink ortamında bir model oluşturulmuştur. MATLAB ve Simulink, görüntü işleme algoritmalarını hızlı ve etkili bir şekilde prototiplemek ve test etmek için ideal bir platform sağlar. Histogram eşitleme modeli, farklı görüntü türleri üzerinde test edilmiş ve performansı değerlendirilmiştir.

Devamında, FPGA üzerinde gerçek zamanlı uygulama için tasarım yapılmıştır. FPGA tabanlı uygulamalar, yüksek hız ve paralel işlem yetenekleri nedeniyle gerçek zamanlı görüntü işleme için idealdir. Histogram eşitleme modeli, FPGA üzerinde donanım tanımlama dili (HDL) kullanılarak uygulanmış ve performansı FPGA platformunda değerlendirilmiştir.

Bu çalışma, bir FPGA üzerinde resim iyileştirme için histogram eşitleme tekniğinin gerçek zamanlı uygulanması için bir metodolojinin ana hatlarını çizerek ve bunu MATLAB-Simulink ortamında modelleyerek sona ermektedir. Histogram eşitlemeyi gerçekleştirmek için model tabanlı bir tasarım tekniği kullanıldı; daha sonra bu modelden oluşturulan SystemVerilog donanım tanımlama kodunu sentezlemek ve simüle etmek için Vivado yazılımı kullanıldı. Bu yaklaşım, görüntü kalitesini iyileştirmek için kullanışlı bir araçtır ve görüntü işleme uygulamalarına uygulanabilir.

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1.INTRODUCTION

In recent years, the emergence of digital image processing techniques in various fields, including astronomy, biology, medicine and geology. Thanks to recent technological advances, digital image processing has become a routine task now required to solve problems in a wide variety of applications and devices. To get the best solutions, the user needs to understand the single problem he has and know how to use the tools available in the market. Image processing is necessary to reduce the size or increase the quality of digital images in order to send or store them. Techniques applied to digital images are used to improve the quality or facilitate the acquisition of information. These studies that increase the quality and usability of the image are called image enhancement. In this study, some of the image enhancement techniques are examined[1].

There are a variety of uses for image enhancement in various industries. Image enhancement is a computational process that requires repeatedly applying a specific function. Using a general-purpose microprocessor or digital image processing is a way to accomplish demanding tasks affordably. Image enhancement techniques involve procedures to highlight or sharpen image elements such as image edges, boundaries, or contrast to increase the usefulness of an image in a graphic. All important areas of image processing require images of such high contrast and visual quality. The application of development techniques does not increase the internal information content, but expands the dynamic range of selected elements. Classifying image enhancement techniques has become very important due to the increasing amount of research conducted in this field [2].

FPGA, or field programmable gate arrays, is therefore a potentially useful tool in the field of digital image processing. The advantages of FPGA technology are reasonable cost and low power consumption. One of the most important reasons for choosing FPGA is that it has a more flexible structure than ASIC (Application-Specific Integrated Circuit) chips and the ability to be programmed over and over again. Thanks to these advantages, it was decided to use FPGA in integrating the model base design into real life [3].

The goal of this project is model based design of image enhancement techniques for FPGA implementation. Rather than use hardware description languages to simplify the design, hardware design was created using MATLAB/Simulink, a model-based design program, along with HDL Coder and Vision HDL Toolbox, which are high-level synthesis tools.

1.1 Literature Survey

Several methods and strategies for image enhancement have been examined in the literature from the standpoint of FPGA implementation. Below is an overview of related projects.

In 1998, “Realtime image histogram equalization using FPGA” is published. Just tiny image sizes are examined in this work; huge images, which would have lengthy processing times and cannot be conveniently stored on the chip, may require additional adjustments [4]. In 2014, “Energy-efficient histogram equalization on FPGA” was published. A limiting system is being created for larger image sizes to reduce the amount of power that External Dynamic Random Access Memory (DRAM) needs to calculate. Realistic frame rates of more than 30 frames per second are achieved using pipeline and data hazard reduction [5].

In 2006 , "A real time contrast enhancement algorithm for infrared images based on plateau histogram” was published. The goal of this project is to use plateau histogram equalization to create an FPGA-based self-adaptive enhancement system for infrared images. The suggested method may significantly increase the contrast of infrared mages and is implemented using FPGA[6].

In 2009 , "FPGA Implementation of High speed Infrared Image Enhancement" was published. In this research, the Successive Mean Quantization Transform (SMQT) is used to automatically enhance infrared images using FPGA implementation. The SMQT preserves the basic shape of the histogram by performing a nonlinear stretch[7].

In 2010, "An FPGA implementation of real-time retinex video image enhancement" was published. In this study, a real-time video image enhancement system was created using FPGA. The proposed system can process color images up to 1900 x 1200 pixels at 60 frames per second using the variational model of retinex theory [8]. In the same year, a new approach for real-time histogram equalization with FPGAs was introduced. Although they create effective structures for histogram equalization, their methods often produce inadequate images.

In 2010, "An FPGA based Real-time Histogram Equalization Circuit for Image Enhancement" was published. This study presents a real-time histogram equalization technique based on FPGA technology to improve visual vision quality. The suggested system computes equalization and histogram statistics by using switching decoders and counters, which are implemented on Altera's FPGA Stratix II series processor.

In 2011, "FPGA Implementation of Image Enhancement Algorithm" was published. It is mentioned how various image enhancement methods such as contrast expansion, brightness

control and histogram equalization are implemented in the FPGA. FPGA is used to implement the algorithms. The findings of the hardware implementation show that optimization strategies perform faster. [9] In 2010, "An FPGA based Real-time Histogram Equalization Circuit for Image Enhancement" was published. To enhance the image's visual quality, it provides real-time histogram equalization based on FPGA technology. The suggested system computes synchronization and histogram statistics by utilizing counters and switching decoders built on Altera's FPGA Stratix II series device [10].

In 2014, “Energy efficient histogram equalization on FPGA” was published. To reduce energy consumption, this project creates an energy-efficient histogram equalization architecture and proposes a memory activation scheme. They design an effective buffering and capping scheme for higher image sizes to minimize the power computation of external Dynamic Random Access Memory (DRAM). Achieving a realistic frame rate above 30 frames per second requires using pipelines and reducing data exposure [11].

2.METHOD

2.1. Image Processing

The scientific and engineering discipline of **image processing** focuses on using computers to analyse, improve, and understand digital images. To increase the meaning of visual data, this field uses a variety of algorithms, including edge detection, segmentation, noise reduction, sharpening, and histogram equalization. In industries including artificial intelligence, automotive, space, health, and security, image processing techniques are commonly employed for tasks like face detection, object recognition, and quality control. Digitizing the image is usually the first step in the process, which then involves transforming it into a format that can be understood using a variety of mathematical techniques [1].

2.1.1 Digital Image

The definition of an image is a two-dimensional function of light intensity, denoted as f (x, y), where x and y stand for the spatial coordinates and the intensity of the image is proportional to the value of f at each point (x, y). An image in digital format can be represented as a matrix where the row and column indices represent the light intensity at each point in the image. An array's constituent pixels are each corresponding elements in the image [1].

2.1.1.1 Classification of Digital Image

There are numerous techniques to classify digital images. The two fundamental types of images should be bitmap and vector. Vector graphics are made up of lines that each provide a solution to a mathematical equation. The stroke coordinates used to make this type of image are precisely calibrated. One of the disadvantages of vector graphics over bitmaps is their lack of detail. The advantage is that you can grow and shrink without compromising quality because the lines are redrew when resizing [1].

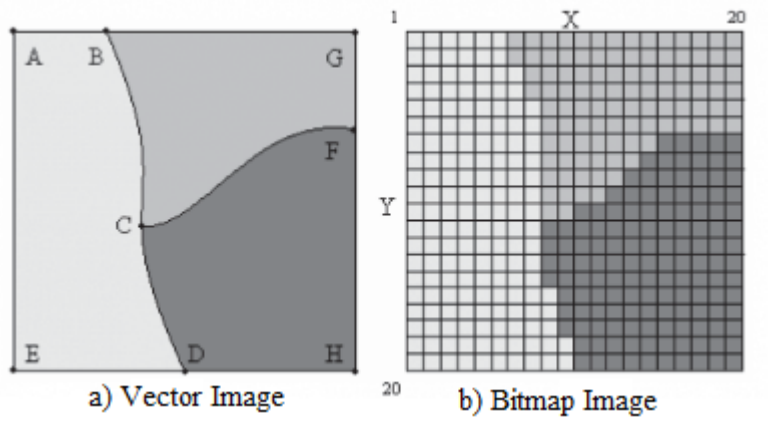


Figure 2.1: Vector and Bitmap İmages [12]

This project makes use of bitmap images and use grayscale and color images to evaluate algorithms within bitmap images, which can be classified as 2D or 3D and have different palettes. Figure 1 illustrates the distinctions between vector and bitmap images.

2.1.2 Color space

When applied to physical devices, color spaces—predefined color ranges—allow for the depiction of colors both digitally and analogically. A color model is an abstract mathematical model that explains how colors can be expressed as tuples of integers.[13] While the model comes in a number of color variations, only the three are important for this project[14].

2.1.2.1 Grayscale

The intensity scale of a digital image, sometimes called monochrome or grayscale level, is a set of MxN values where each pixel is a single sample containing information about

the intensity of the image. The brightness value of each pixel in a grayscale image varies between 0 (black) and 1 (white). Up to 256 shades of gray are typically available in this mode (8 bits per sampled pixel). When the bit value changes, the grayscale range of the pixel also changes[14].

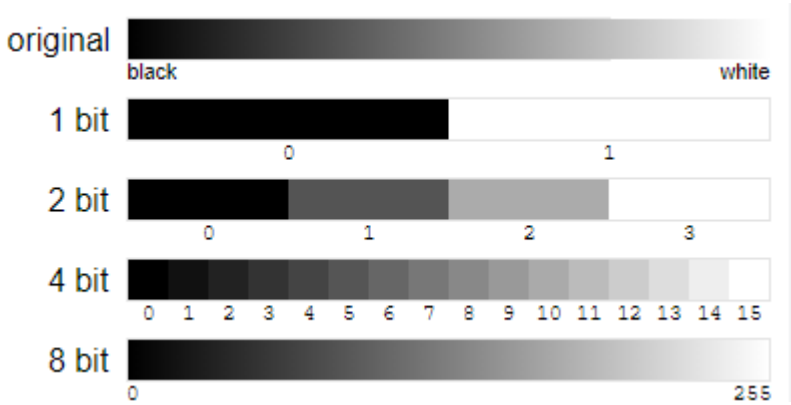


Figure 2.2 : Grayscale levels of images [15]

Hue (color), value (lightness or darkening), and saturation (purity of color) are the three qualities that can be used to characterize a color. Therefore, there is no one proper way to convert a color image to grayscale; instead, the most popular method is to keep the brightness information while removing the hue and saturation values. Given the supposition that red, green, and blue represent light, equation (1) can be used to approximate a grayscale image from a color image. Here, 0 denotes less intensity, or black, and 1 denotes more brightness, or white. The relative brightness of red, green, and blue light as seen by the average individual is fairly accurately represented by this formula[14].

$$Grayscale = 0.299 * Red + 0.587 * Green + 0.114 * Blue \qquad (1)$$

2.1.2.2 Red-Green-Blue Model

An array of 3xMxN pixels, each of which represents the red, green, and blue components of a color image, is referred to as an RGB image. Red, green, and blue are the three color inputs that make up the RGB model, which can be thought of as a stack of three scale image intensities to be displayed on a color monitor. Primary hues are defined as red, green, and blue. Typically, a scale from 0 to 255 is used to assess the intensity of each component (one byte for each component)[1].

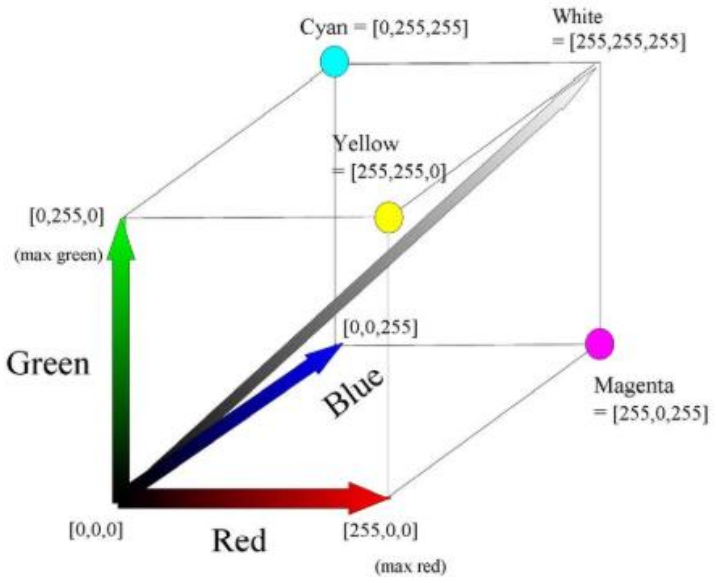


Figure 2.3 : The RGB color cube schematic[1]

This model is crucial to image processing since it is the most often used to show digital images in the formats that are now available on a screen.

2.1.2.3 Hue-Saturation-Brightness Model

The human sense of color serves as the foundation for the HSB(Hue-Saturation-Brightness) model. There are three different categories of parameters. Hue is the quality of a visual perception that makes a scene appear to resemble one of the four recognized colors (red, yellow, green, or blue) or a mixture of two of them. Saturation is the area's degree of color measured in relation to its brightness. The quality of a visual perception that determines how much or how little light an area appears to be emitting is called brightness[1].

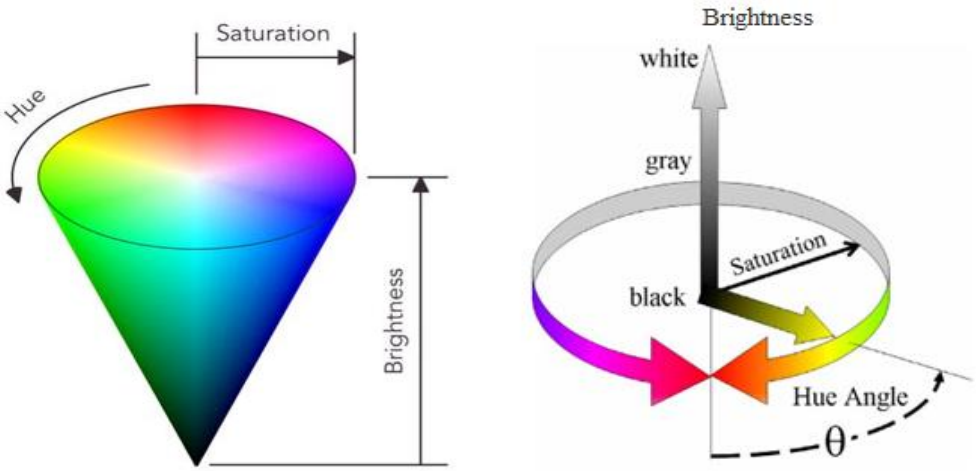


Figure 2.4 : Coordinate system for “Hue, Saturation, Brightness (value) [1]

The RGB (Red-Green-Blue) model serves as the foundation for the HSB color model, which uses cylindrical coordinates. The following are the ways in which these components are used in cylindrical coordinates:[16}

• H (hue) is typically expressed as a circumference, meaning that the degree indicates the color of that pixel; but, in certain situations, it can also be used as a percentage.

• S (saturation) the distance from the cylinder axis is the depiction of this component.

• B (brightness), also known as V(value). Typically, it is shown on a scale of 0 to 1. Because a pixel with a value of 0 is always black, independent of the other two components, the HSB model can also be understood and seen as a cone .

The following provides the RGB to HSV conversion: [16}

$$H = \begin{cases} Undefined, & MAX = MIN \\ 60° * \frac{G-B}{MAX-MIN}, & MAX = R \text{ and } G > B \\ 60° * \frac{G-B}{MAX-MIN} + 360°, & MAX = R \text{ and } G < B \\ 60° * \frac{B-R}{MAX-MIN} + 120°, & MAX = G \\ 60° * \frac{R-G}{MAX-MIN} + 120°, & MAX = B \end{cases} \tag{2}$$

$$S = \begin{cases} 0, & MAX = 0 \\ 1 - \frac{MIN}{MAX}, & Otherwise \end{cases}$$

$$V = MAX$$

As a result, the formula shows that the Value or Brightness of a pixel is the highest value of any RGB component.

2.2 Model Based Desing

The traditional design process and model-based design are very different. With continuous-time and discrete-time building blocks, designers can utilize model-based design to define plant models with advanced functional qualities instead of relying on intricate

architectures and large amounts of software code. Software testing, verification, and fast prototyping can be facilitated by utilizing these constructed models in conjunction with simulation tools. Along with improving testing and verification, the new design paradigm sometimes makes it possible to test dynamic effects on the system much faster and more effectively than with the previous design methodology by utilizing hardware-in-the-loop simulation[17].

At present, Model Based Design is a well-known hardware methodology that utilizes an extensive array of specialized software tools, including MATLAB/Simulink [18]. A common platform for the integration of different stages of the development process is provided by this design approach [19]. In addition, Model Based Design optimizes the models to meet speed, space, and power restrictions by automating the creation and verification of HDL code using MATLAB/Simulink.

So, quick prototyping, system validation, and testing can be accomplished with Model Based Design in an efficient manner. MATLAB/Simulink's HDL Coder, which facilitates automated VHDL code creation for FPGA implementation, is utilized for the quick design of picture enhancement. The image enhancement method can be immediately passed to an FPGA implementation through the use of the HDL Coder in Model Based Design flow[18].

2.2.1 Model-Based Design Process

System engineers utilize Model-Based Design to connect with suppliers and customers by using models to extract low-level needs. They simulate design trade-offs and simulate the behaviour of the system's digital, analog, and radiofrequency components—tasks that can be challenging or impossible to accomplish with paper-based design assessments. The same models can then be expanded upon by algorithm developers to create and test more intricate designs. These models may eventually serve as the design artifacts from which hardware designers automatically produce HDL code during the development process. The ended hardware and HDL implementation's performance can then be compared to the model-level findings using the system-level models and tests as a test bench. Model-Based Design minimizes the amount of translation errors and inefficiencies by reusing and expanding on models at every stage of development. Design iterations are quicker, design artifacts are automatically generated, and the engineers have a single platform to communicate design when the model is at the center of the process [20]. Usually, modelling any system involves these steps ; determining the components of the system, defining the system, using equations to model the system, constructing the model ,executing the simulation ,verifying the simulation outcomes.

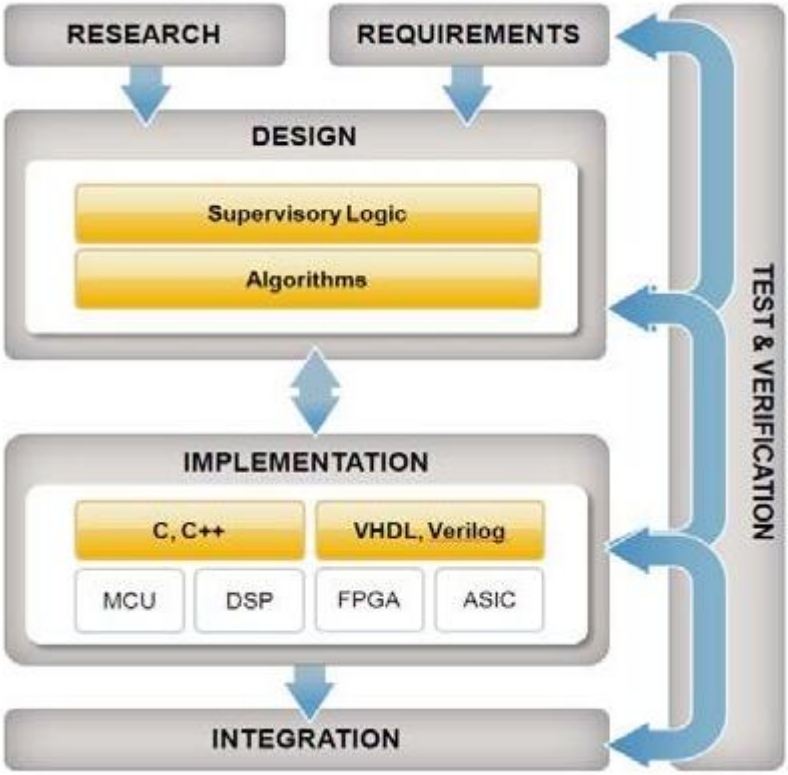


Figure 2.5 : Stages of the model-based design process [20]

The initial three phases of this method are carried out outside of the software environment before the model is developed. These initial three steps are crucial, and many individuals tend to make mistakes in them. Ultimately, building the model is a relatively standard process once the system has been represented by equations. One could argue that since no model is flawless, certain details should always be left out and the system description should be made simpler. Experience is needed for the entire process. A very complex model and a slow simulation are the outcomes of attempting to create an ideal model [21].

2.2.2 Model-based design Advantages/Disadvantages

When compared to the conventional method, model-based design has the following advantages: [22]

- Model-based design gives different (development) groups a shared design environment that makes general communication, data analysis, and system verification easier.
- Early in the system design process, when the time and cost implications of changing the system are at their lowest, engineers can identify and fix problems.

- It facilitates the reuse of designs for upgrades and for derivative systems with more functionality.

The disadvantages of model-based design are well known in product development and are as follows:[22]

- The general-purpose approach used in typical embedded and systems development might result in laborious porting between processors and ecosystems, outweighing the advantages of more straightforward lab-based solutions.
- There are many fence post mistakes and other typical compilation issues that may be readily fixed in traditional systems engineering, yet a large portion of the compilation tool chain is closed source.

2.3 Field Programmable Gate Array

Real-time image processing applications frequently make use of FPGAs as implementation platforms. An end-user can directly modify the final logic structure of a Field Programmable Gate Array (FPGA) because it is a programmable (or reconfigurable) device [23]. An FPGA is comprised of a range of non-committal elements that can be set, programmed, or linked in a virtually infinite number of ways based on user specifications. The versatility of general-purpose processors and the hardware-based speed of ASICs can be balanced by using an FPGA, which is reprogrammable and readily upgradeable.

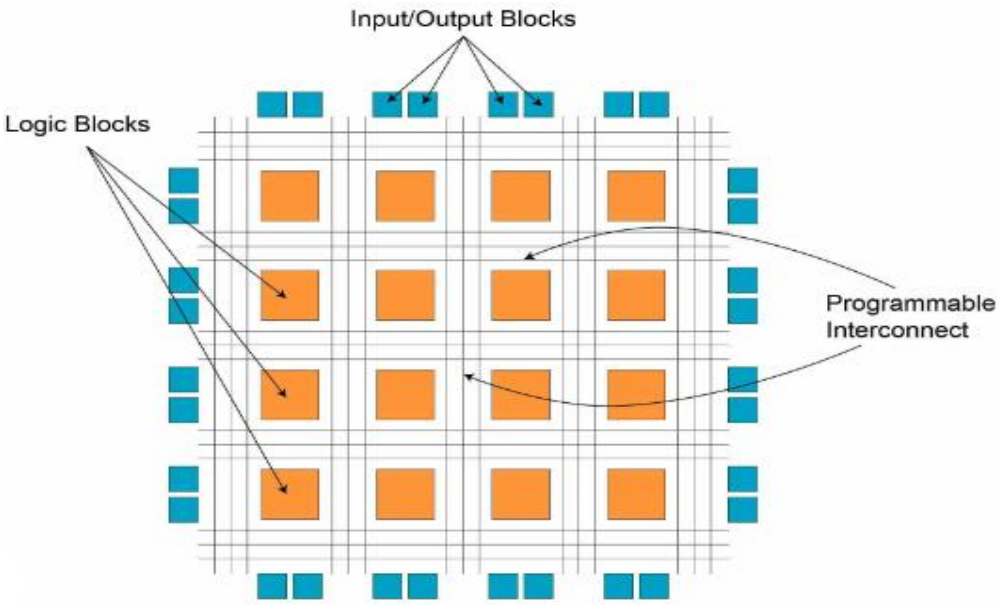


Figure 2.6 : FPGA's Internal Structure[24]

2.3.1 Disadvantage of ASICs and DSPs

Options for system architectures that can be implemented on hardware include programmable solutions like FPGAs and digital signal processors (DSPs), as well as standard cell ASICs and ASSPs. Every one of these has benefits and drawbacks. The development of Application-Specific Standard Product ASSPs requires a lot of money, effort, and flexibility. The best performance is provided by a fully customized ASIC design, but it cannot be altered. Therefore, every hardware design flaw that arises after production wastes the entire product. Specialized microprocessors known as DSPs are usually programmed in C or, for better performance, assembly code. It works well for really complicated math-intensive applications, such image processing. Although hardware design still requires some knowledge, it has a far flatter learning curve than other design options. Nevertheless, strong DSPs can be expensive, and the software that goes with them might not be as good as the hardware. FPGAs are perfect for real-time video processing because of their numerous additional characteristics and programmable computing technologies [25]. An FPGA can be used to explore hardware design approaches like pipelining and parallelism , which is not achievable with dedicated DSP designs. The MIPS needs of an application are the main factor that influences most engineers' decision to choose an FPGA rather than a DSP [23]. Processor-only designs are not acceptable due to the requirement for high performance. The only programmable devices that can solve this issue are FPGAs.

2.3.2 Advantages of FPGAs

Many advantages of FPGAs make them a preferred choice of implementation in DIP realm[25].

- Many image processing techniques, such as the four-stage edge detector, need numerous iterative data set processing. By combining these stages into a single run, FPGA can execute many image windows and operations concurrently by taking use of temporal and spatial parallelism.
- Several optimization approaches, like Loop Fusion and Loop Unrolling, can be utilized to optimize FPGA resources efficiently and accelerate implementations by reducing the number of redundant processes.
- The ability to conduct read, process, and write simultaneously is a feature of FPGAs' parallel I/O capabilities.

- An FPGA can be completely rebuilt using a new design. When the device can be reprogrammed, a new logic design can be installed in its place, depending on the capabilities of the processor and allowing for a wide range of design options.
- In the development of image processing, the reusability and efficiency of FPGA-implemented hardware are extremely helpful since they enable an effective system in terms of both cost and performance. The ability to quickly integrate IP blocks without requiring changes or repeating the verification cycle makes troubleshooting easier and cuts down on time-to-market significantly.
- Even the lowest volume consumer market may afford the development expenses of FPGAs thanks to their high computational density and cheap development costs.

2.3.3 FPGAs' Limitations

Although they have certain drawbacks, FPGAs are a powerful and exciting platform for real-time image processing applications. They are only appropriate for algorithms that do not require complicated mathematics or floating-point, and they have overheads such as data transfer times and reconfiguration times. FPGAs require a lot of work to modify or combine circuits, therefore reconfiguring them quickly is also not possible. Standard logic cells have a limited capacity for implementing memory, and procedures that cannot be divided into smaller jobs need for a more serial processing approach. Hardware increases speed but requires more time to create.

2.3.4 Implementation on FPGAs

FPGAs have been the subject of recent investigation as a DIP algorithm development platform. To represent DIP algorithms, researchers have created the high-level language SA-C. They have also created an optimized compiler that combines SA-C code with other high-level programs to create programs that can be executed on FPGAs. For eight typical DIP routines, FPGAs provide speeds that are 8–800 times faster than Pentium CPUs. The key characteristics of FPGAs, especially in image processing, are highlighted by experiment results and analysis of problems with pipelining, parallelism, optimizations, memory, and I/O. An FPGA-implemented pipelined architecture for image processing techniques, including edge detection, convolution, median filter, and fundamental morphological operators, is presented by the authors[26].

2.4 Histogram Based Image Enhancement

The technique of altering digital images to make the results more appropriate for display or additional image analysis is known as image enhancement. Digital image processing, involves using a computer to alter digital data in order to improve the quality of the image. Processing aids in optimizing the clarity, precision, and details of features that are important for information extraction and subsequent analysis. For instance, it can brighten, sharpen, or eliminate noise from an image to make it easier to see the important details .

İmage enhancement methods are especially helpful because many images do not provide enough information for image interpretation when viewed on a color display. There's no deliberate attempt to enhance the image's fidelity in relation to a perfect version of the image. There are many different methods available for enhancing image quality. The more widely used methods are edge enhancement, spatial filtering, density slicing, and contrast stretch. After the image has been adjusted for geometric and radiometric errors, an attempt is made at image enhancement. Every band in a multispectral image receives a different application of image enhancement techniques. For image enhancement, digital procedures have been proven to be more satisfactory than photographic techniques due to their vast range of digital operations and precision.

Gray level pixel modifications can be used to enhance images at a fundamental level. Several spatial domain and frequency domain filters have been developed in the past to reduce noise in photographs. Histogram equalization is an important technique in spatial methodologies for image enhancement schemes. Frequency domain and spatial domain filters

are typically hard to implement in hardware for low power and high speed criteria due to their computational complexity. Because of this, Histogram Equalization (HE) techniques are crucial, and research has shown that they can significantly increase image quality at a tolerable computing cost. The investigation into employing diverse ways to enhance image contrast is still ongoing. Nevertheless, the quality of the picture taken affects how effective the methods are. There are several methods for improving images, and histogram equalization is one of the most widely used ones [27].

In this research, Histogram equalization was chosen among the image enhancement techniques.

2.4.1 Histogram Equalization

The process of increasing an image's histogram's range of motion is called histogram equalization. By assigning the intensity levels of each pixel in the input image to the histogram, equalization creates an output image with a uniform distribution of intensities. The objective of histogram equalization is to achieve a homogeneous histogram while also enhancing contrast. This method can be applied to the entire image or only to a specific area. Intensity distributions are redistributed by histogram equalization. An image with a lot of peaks and valleys in its histogram will still have hills and valleys after equalization, but they will be moved [28].

A technique called histogram equalization was developed to maximize the dynamic range of any given image. This method involves calculating each level's likelihood and then reassigning a new level in accordance with that probability. Since the data are discrete, the outcome should ideally be a flat histogram with equal probabilities for each level. However, actual findings show that this is not the case. There will be greater contrast in the updated image compared to the original.

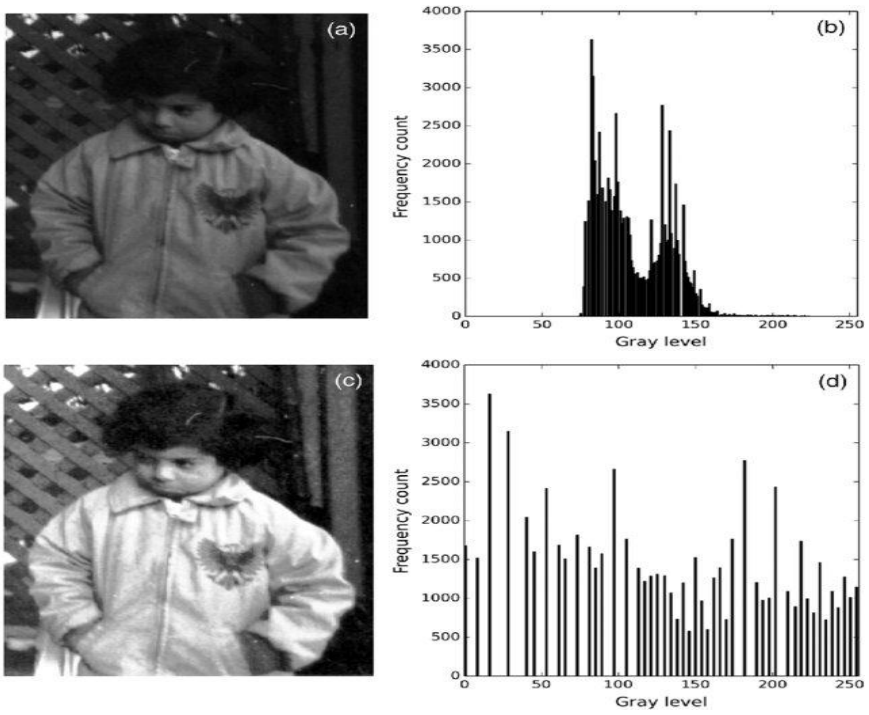


Figure 2.7 : Grayscale image. Original and after Histogram Equalization [29]

When the image's useful data is represented by close contrast values, this strategy typically improves the global contrast of numerous photos. The intensities on the histogram can be more evenly spread by making this adjustment. This makes it possible for regions with less local contrast to become more contrasty. This is achieved by efficiently dispersing the most frequent intensity values by histogram equalization.

The technique works well in pictures where the foreground and backdrop are either completely dark or completely light. Specifically, using this strategy can result in enhanced views of the bone structure in x-ray images and improved detail in overexposed or underexposed images. The method's invertible operator and relatively simple approach are two of its main advantages. The original histogram should therefore be recoverable in theory if the histogram equalization algorithm is known. The computation doesn't require a lot of processing power. The approach has the drawback of being indiscriminate. It might reduce the useable signal while raising the contrast of background noise[30].

2.4.1.1 Histogram Equalization Techniques

The intricate techniques of Histogram Equalization (HE) are applied in order to improve the contrast of the photographs. This paper discusses a few HE approaches.

- **Histogram Equalization (HE)**

A method for enhancing contrast in images is called histogram equalization. An picture's histogram primarily shows the relative frequency with which the different gray levels appear in the image [31].

The efficacy of the Histogram Equalization (HE) approach is impacted by two primary drawbacks [32]. One gray level is divided into two distinct neighbour gray levels with varying brightness using histogram equalization (HE). Histogram Equalization (HE) assigns a gray level with higher intensity to an image if it contains a gray level for the majority of its content, giving the resulting image a washed-out appearance [32]. In order to get the greatest value of the evenly distributed image, HE can start a significant change in the brightness of an image.

- **Global Histogram Equalization (GHE)**

This is a histogram technique that displays both the original and final photos after acquiring the user-provided input image and improving it globally . GHE provides a notable improvement in contrast between images. Images' contrast is resolved in their dynamic range, which is the ratio of the brightness of the brightest and darkest pixels. An image's contrast and general intensity distribution can be found out via the histogram [33].

- **Local Histogram Equalization (LHE)**

Although GHE describes the worldwide data, it is unable to modify the local lighting conditions. The process of block-overlapped histogram equalization is called local histogram equalization, or LHE. LHE extracts the data and classifies a sub-block . Next, the sub-block's center pixel that is utilized in the CDF benefits from the histogram equalization. The sub-block histogram is then repeated after one pixel is stimulated, until the desired image is achieved. LHE is not well suited to partial brightness in sequence; further enhancing of certain sections is contingent upon mask size [33].

- **Adaptive Histogram Equalization (AHE)**

For both natural and medical images, adaptive histogram equalization (AHE) is an excellent contrast enhancement technique. The process involves applying to every HE pixel . There are two types of histogram equalization: adaptive and non-adaptive. Every pixel in the non-adaptive image is altered using the same computation pattern, which makes use of the entire original image's histogram. In studies that produce superior results for pictures with features obscured by dark areas [34]. Every pixel in the adaptable is altered according to the pixels in the vicinity of that pixel.

In this project, the general histogram equalization technique, namely Histogram Equation (HE), is used.

2.4.1.2 Contrast Enhancement

An image's contrast, which is commonly defined as the variation in brightness or gray level values, is a crucial feature. It can be described as the proportion of an image's maximum to minimum intensity. An image's capacity to identify and resolve details is significantly

influenced by its contrast ratio. The image can be interpreted more easily the larger the ratio[35].

Contrast enhancement techniques increase an image's brightness value range, allowing the image to be efficiently shown in the analyst's preferred manner. A scene's density values are essentially stretched throughout a wider range by being moved farther apart. Increased visual contrast between two areas with varying uniform densities is the result. This makes it simple for the analyst to distinguish between areas with initially slight variations in density.

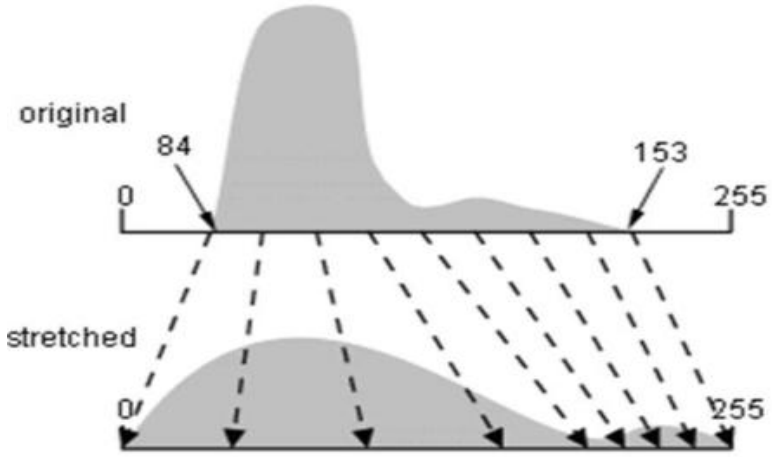


Figure 2.8 : Contrast Stretching [36]

The data values used in these approaches undergo a non-linear transformation in both the input and output. With x representing the input data value and y representing the output data value, y = f (x) defines the general form of the non-linear contrast enhancement. It has been discovered that the nearly classes and subclasses of a primary class have better color contrast when using non-linear contrast enhancement approaches. One kind of nonlinear contrast stretch is a logarithmic scaling of the input data. The histogram's darker regions' brightness values are more affected by this improvement. By scaling the input data with an inverse log function, it is possible to reverse it and increase values in the brighter region of the histogram[35].

2.4.1.3 Cumulative Histogram

The process of histogram equalization involves employing the cumulative histogram as a look-up table or transfer function to uniformly redistribute the pixel intensity values.

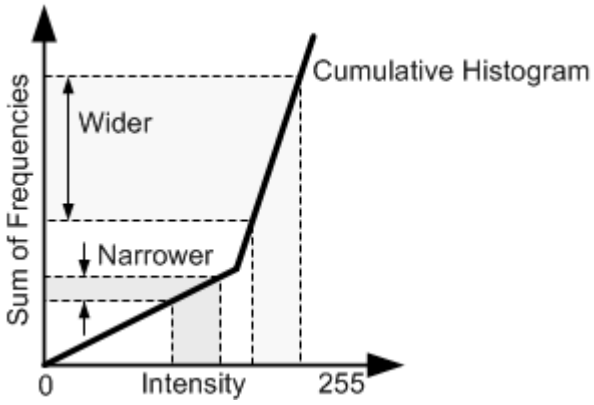


Figure 2.9: Slope Of Cumulative Histogram [37]

It is from the cumulative histogram that the concept of histogram equalization originates. View the example cumulative histogram on the image. Each intensity value is associated with an accumulation of frequencies (pixel count). Greater steepness corresponds with higher pixel density. One interval has a slow steepness, and the other has a steep slope. Both intervals have the same width, but they have different slopes. When map the intensity values' intervals to the other axis, they exhibit distinct intervals on the projected axis after projection. The interval narrows if the slope is smaller than 1. The interval will be bigger if the slope is higher than 1. Stated otherwise, in the case of a high pixel density, expanded the intensity range; in the case of a low density, is decreased the interval. The limited contrast range of an overexposed or underexposed image is ideal for the greatest results from histogram equalization.

2.4.1.4 Color Channel Stretching

When performing histogram equalization on an RGB (Red, Green, Blue) image, histogram equalization can be performed separately for each color channel. However, synchronized histograms can sometimes produce undesirable results, especially color imbalance. Therefore, by using color channel stretching, the contrast or distribution of each color channel can be more finely adjusted and unwanted color imbalances can be avoided.

Taking into consideration the input color image I(u, v) {R(u, v), G(u, v), B(u, v)}, in which {R, G, B} denotes the red, green, and blue color spaces and (u, v) represents the pixel coordinates. In order to adjust for any color distortion resulting from an unfavourable capturing situation, the input image is first stretched on each color channel. Stretching improves the color content of the image and prepares it for contrast improvement. All color

signals are transformed to the Hue–saturation–intensity (HSI) space after magnitude stretching. The system processes the I-signal, which indicates brightness or intensity. Since the H-channel and S-channel components have no direct effect on how contrast is perceived by humans, they are left unaltered. However, because the human visual system is so sensitive to variations in intensity, the intensity I-channel is processed during the augmentation phase [38].

3. HISTOGRAM EQUALIZATION DESIGN, SIMULATION AND SYNTHESIS

3.1 Histogram Equalization HDL

A traditional image processing method for improving a picture's overall contrast is called histogram equalization. This technique redistributes the grayscale values to more evenly divide the intensity levels across the available range after analysing the full image's histogram in a single step. The input image is split into four parts in the example created with the Vision HDL Toolbox, and parallel histogram calculations are carried out. After that, a global equalization is applied to the entire image by combining these partial histograms. Due to its simple methodology and very low hardware resource requirements, this procedure is efficient and well-suited for real-time processing on FPGAs or comparable hardware platforms. When the illumination is consistent and the majority of the image details are already quite visible, histogram equalization performs especially well. For example, this technique can equalize contrast and improve the image's perceived sharpness in a well-lit scene taken during the day. But because it affects the entire image worldwide, it could not work well in scenes with different lighting conditions—some areas might appear too bright, while others might lose clarity. Therefore, in low-cost, high-speed systems when simple contrast enhancement is adequate, global histogram equalization is usually preferred[39].

3.1.1 Model Based Design of Histogram Equalization HDL

In this study, histogram equalization consists of three parts. These operations consist of video partition, HDLHistogram and Equalization. A ready-made histogram equalization block was not used in modeling. HDLHistogram and Equalization blocks are designed as separate blocks containing subsystems.

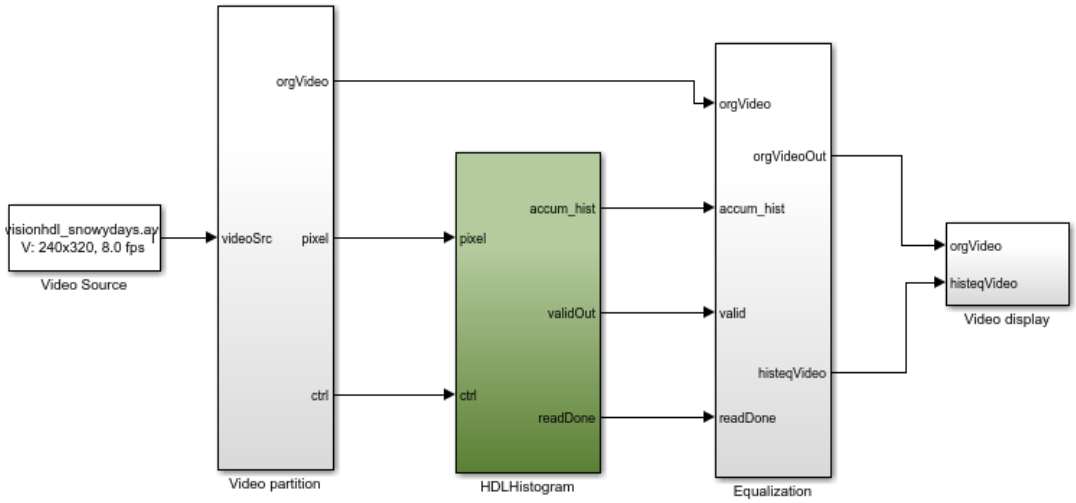


Figure 3.1 Gray Image Histogram Equalization Block Diagram

The histogram can be calculated both on the entire image or separately from small parts of the image. When calculating the histogram of a large image, the image can be divided into smaller pieces. In this diagram, the image coming from the input in the video partition block is divided into 4 equal parts. These divided parts are converted into pixels and control signals of pixels. The pixels and control signals of these 4 parts are combined in common frames. It is sent to the outputs of the video partition block as Pixel and Ctrl.

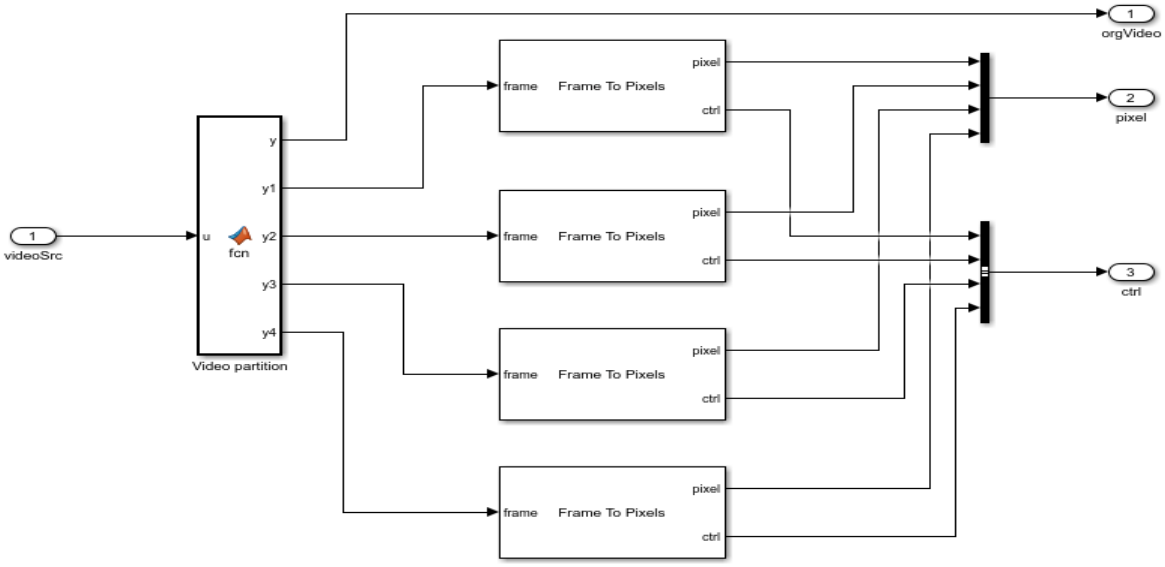


Figure 3.2 Video Partiton Model subsystem

In the HDLHistogram subsystem, histogram calculation is made with the pixels and ctrls coming from the video partition block. Since the input image is grayscale, it consists of 256 separate panes. All these divisions are read in a loop from the histogram block. Then the bin values calculated in the histogram are calculated in the hist_accum block to calculate the

cumulative histogram. The calculated values are sent to calculate Equalization. The result of Equalization is converted to the output image and the output is received.

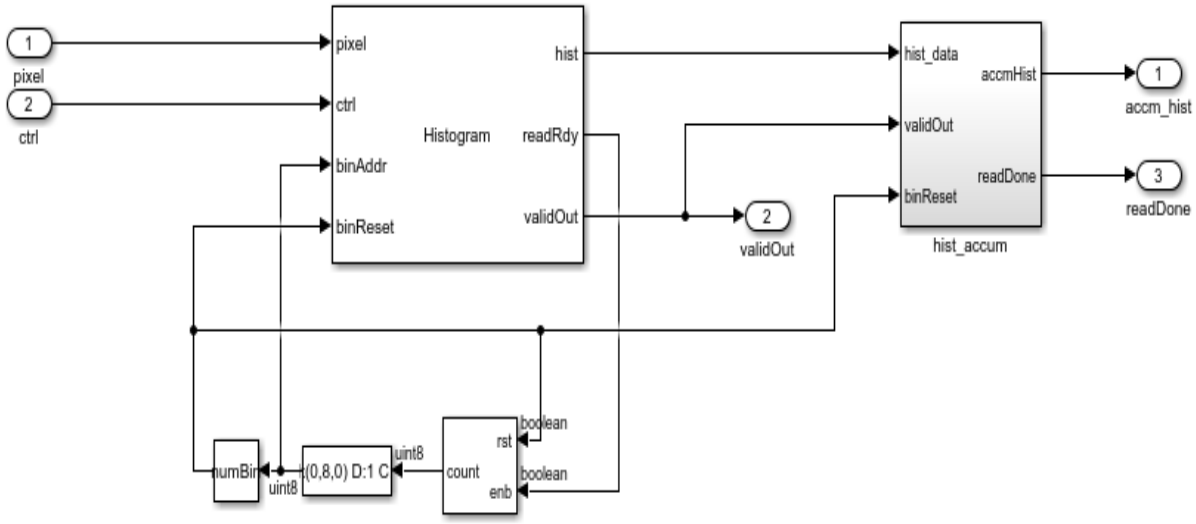


Figure 3.3 HDLHistogram Model subsystem

3.1.2 Simulation in Simulink of Histogram Equalization HDL

In order to observe the results of the histogram equalization project, a video file was added to the Simulink environment using the "Video Display" block. This video is a standard test video that is commonly used in image processing applications for contrast enhancement. The video was opened in MATLAB and modified to have low contrast, making the effect of the histogram equalization algorithm more noticeable. The results can be observed in Figures 1.4(a) and 1.4(b). The obtained results show that the designed histogram equalization algorithm significantly improves the overall contrast of the video. It is clearly seen that details in darker regions become more distinct, and the overall brightness of the video is balanced. If the designer had used a simple brightness enhancement method instead of histogram equalization, contrast loss could have occurred, and some details might have disappeared entirely rather than becoming more visible. Moreover, some basic contrast enhancement filters can also increase noise, leading to undesirable results. Histogram equalization minimizes such issues and generally improves the video quality.

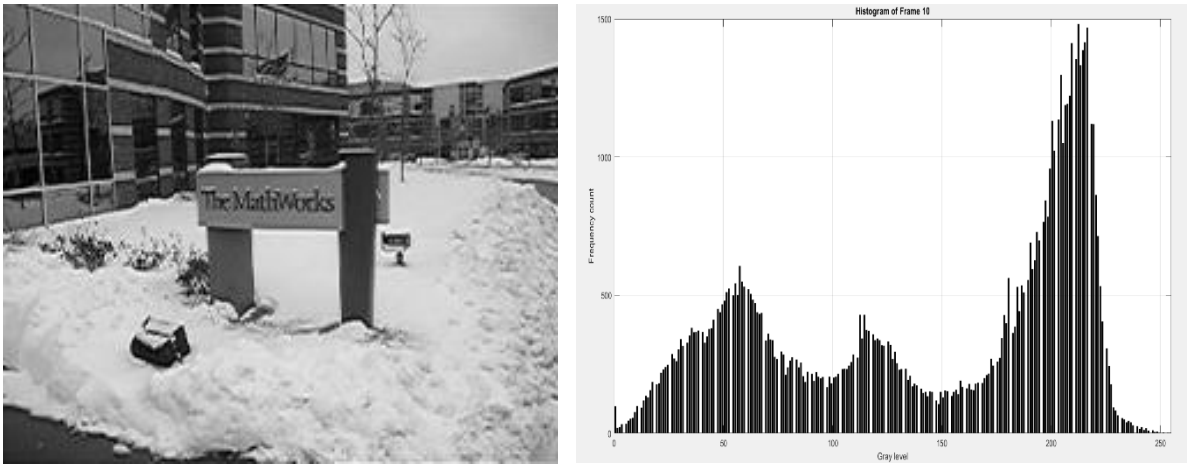


Figure 3.4 : (a) Original Image [39] (b) Histogram Graph of Original Image (c) Result of Image Histogram Equalization [39] (d) Histogram Graph of Result Image

3.1.3 HDL Code Generation of Histogram Equalization HDL

In order to generate VHDL and SystemVerilog code from model-based designs in Simulink, designers need to use the “HDL Coder” toolbox in Simulink. This toolbox includes two main components: “HDL Coder Advisor” and “HDL Workflow Advisor.” First, the design must be validated using the “HDL Coder Advisor.” Once the design successfully passes all checks, the “HDL Workflow Advisor” is then used. The output of the “HDL Workflow Advisor,” which completes all processes, is shown in Figure 1.5.

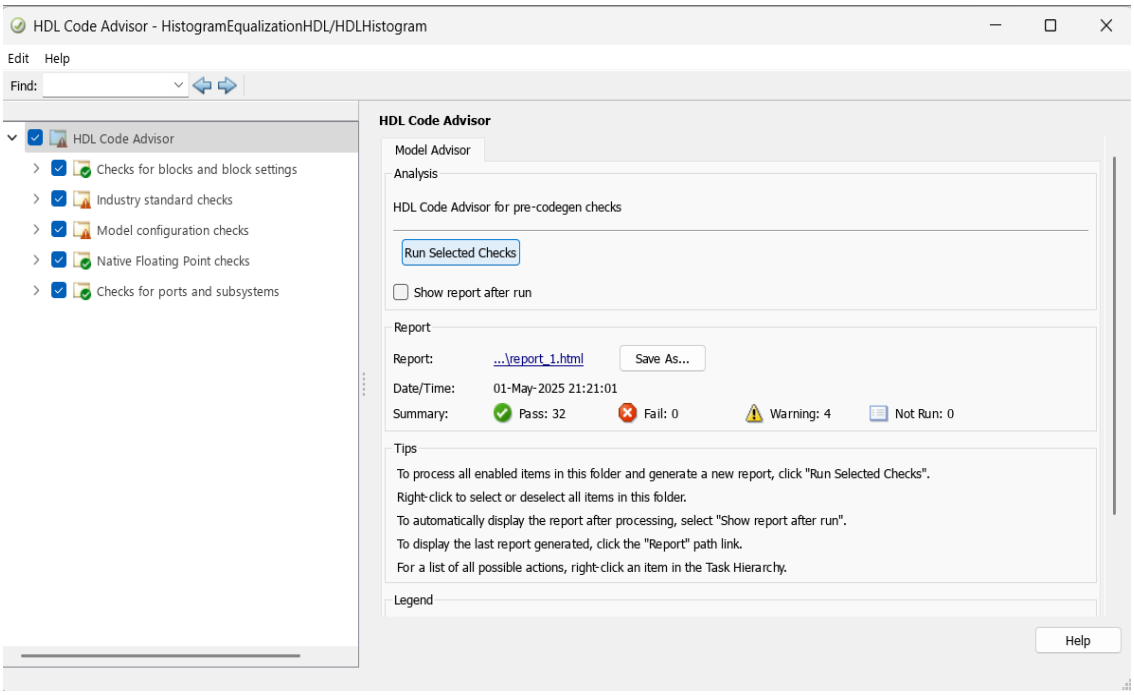


Figure 3.5. HDL Workflow Advisor Result

After testing that the blocks in the Simulink model are compatible with HDL Coder, the HDLHistogram block is right-clicked to generate the HDL code. Then, the HDL code option is clicked, followed by the HDL Coder properties option.

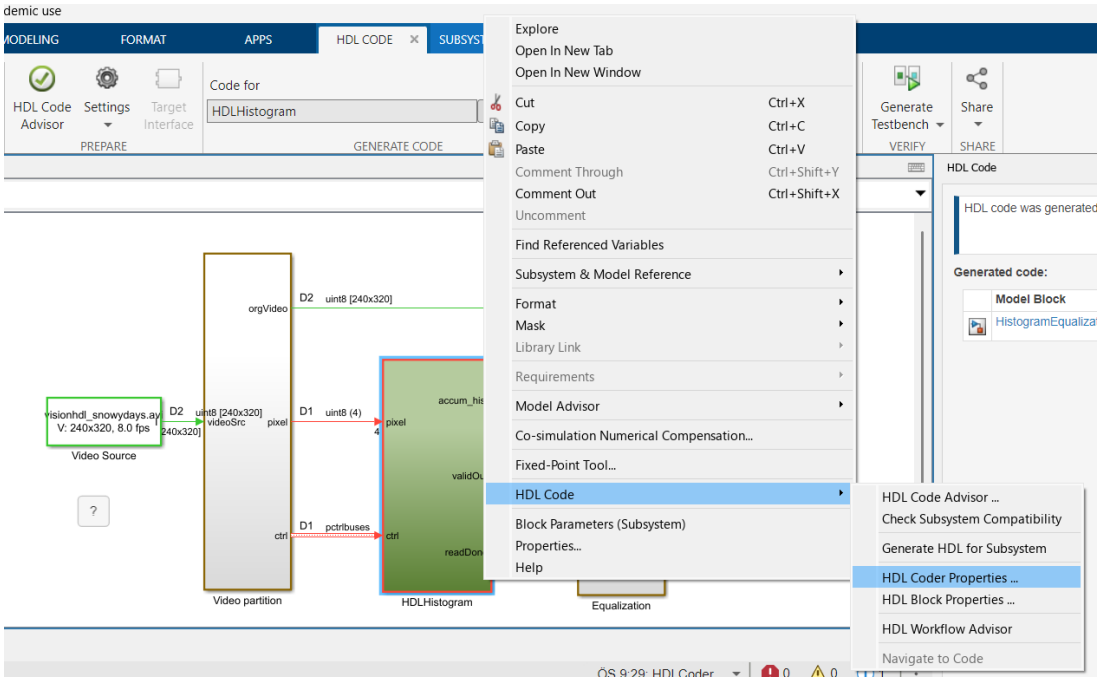


Figure 3.6. HDL coder Properties Selection

From the window that opens, SystemVerilog is selected as the HDL code language to be generated. Then, OK is clicked to generate the code. The code is generated and prepared ready for Vivado.

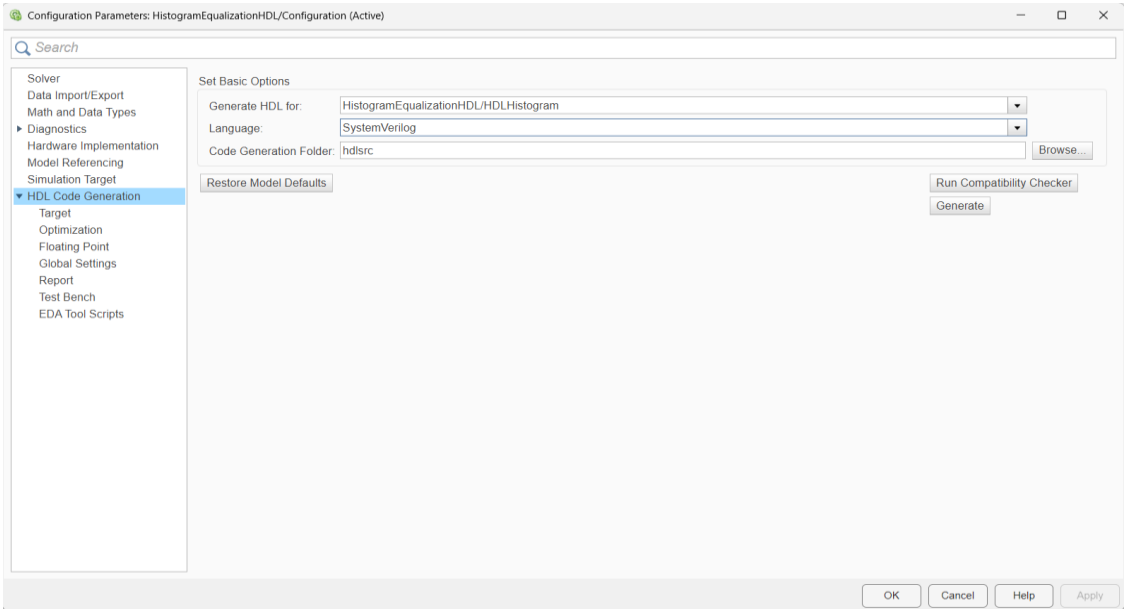


Figure 3.7. HDL coder Generation window

After the HDL code of the system is generated, a suitable testbench code needs to be created. For this, the Generate Testbench option in the HDL Coder application linked to MATLAB Simulink is selected to generate the testbench code.

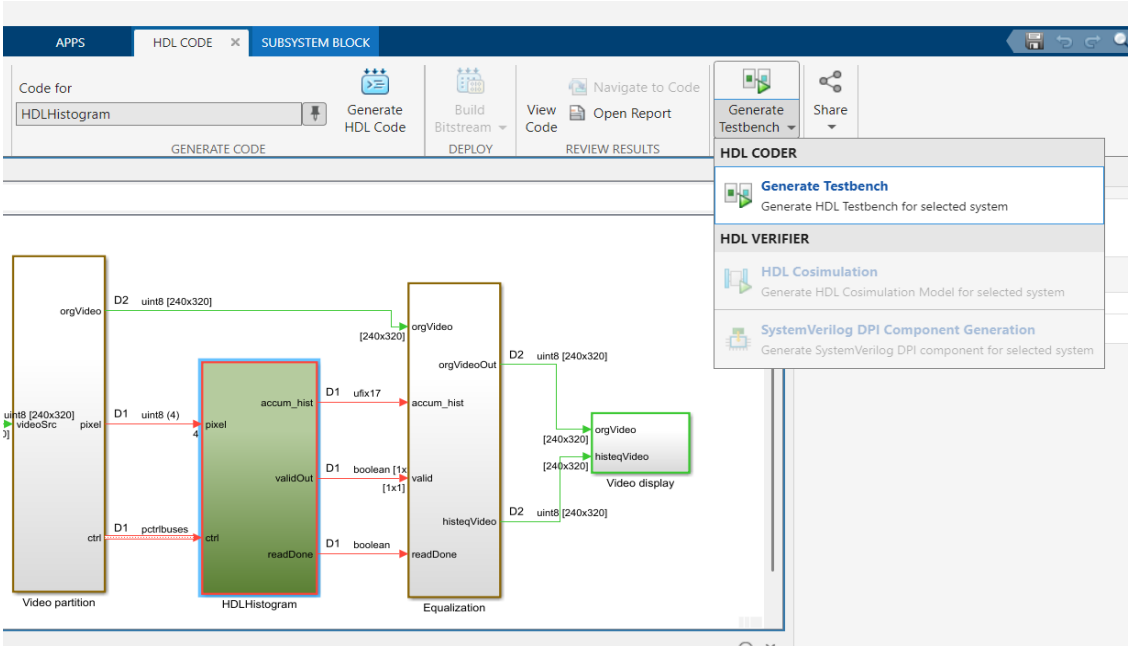


Figure 3.8. HDL coder Generate Testbench Selection

3.1.4 Synthesis and Simulation in Vivado of Histogram Equalization HDL

After the SystemVerilog codes of HDL Histogram Equalization were obtained, these source files were added to Xilinx Vivado 2024.2. Then, the “Run Synthesis” option was selected, and the RTL schematic of the codes was generated. The main system and the subsystem of the RTL schematic are shown in Figures 3.9 and 3.10, respectively.

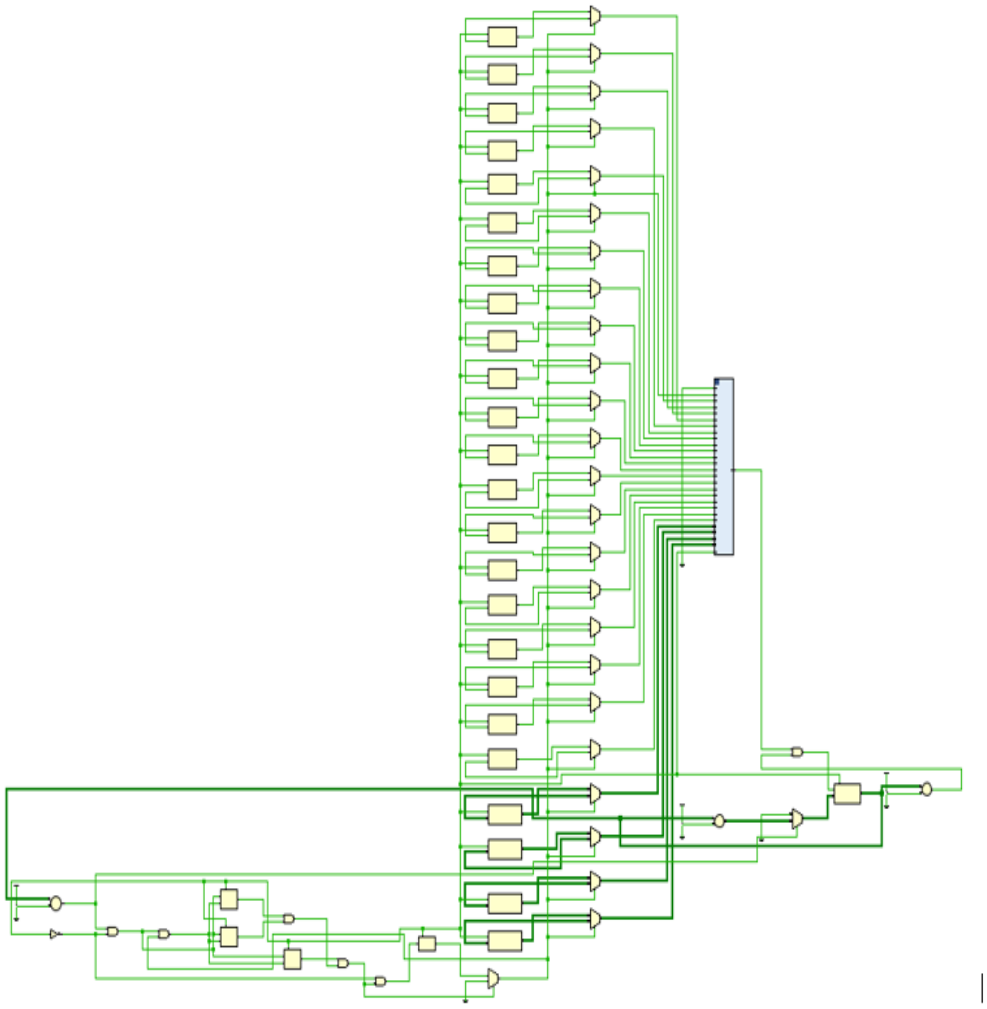


Figure 3.9. RTL Schematic of HDL Histogram Equalization

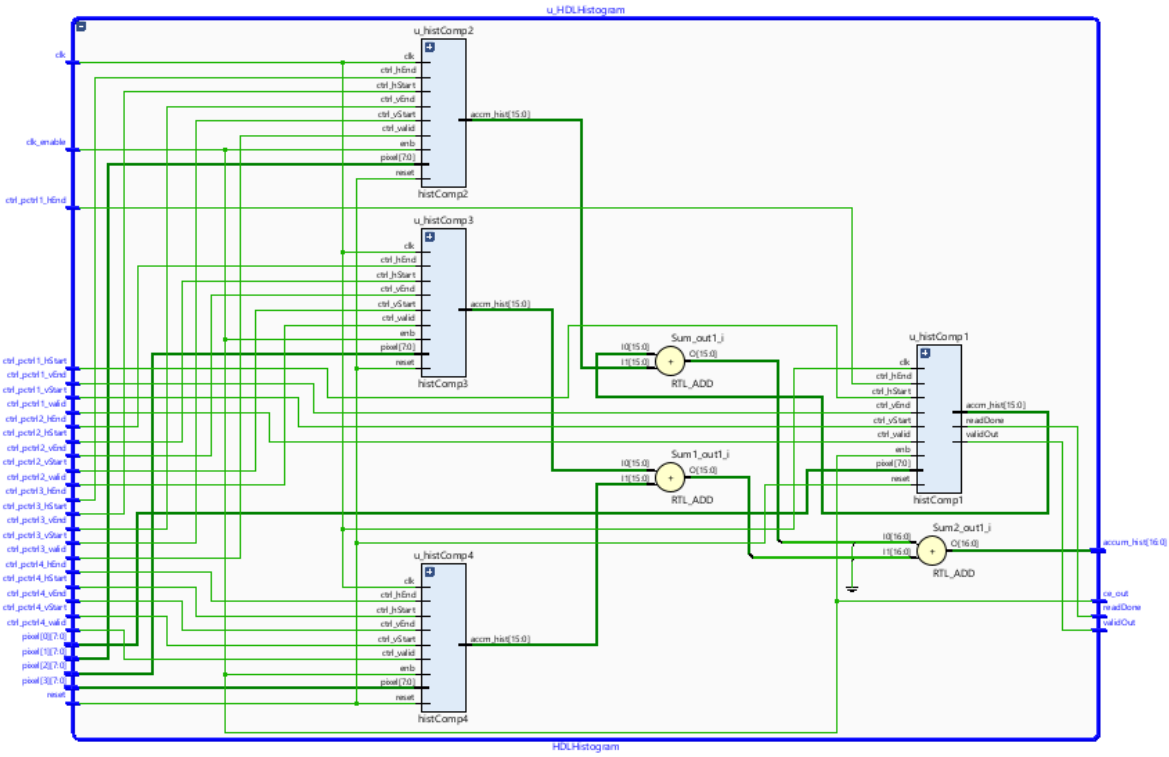


Figure 3.10**.** RTL Schematic Subsystem of HDL Histogram Equalization

Figure 3.11 displays the test results of simulating the HDL Histogram Equalization testbench code in Xilinx Vivado.

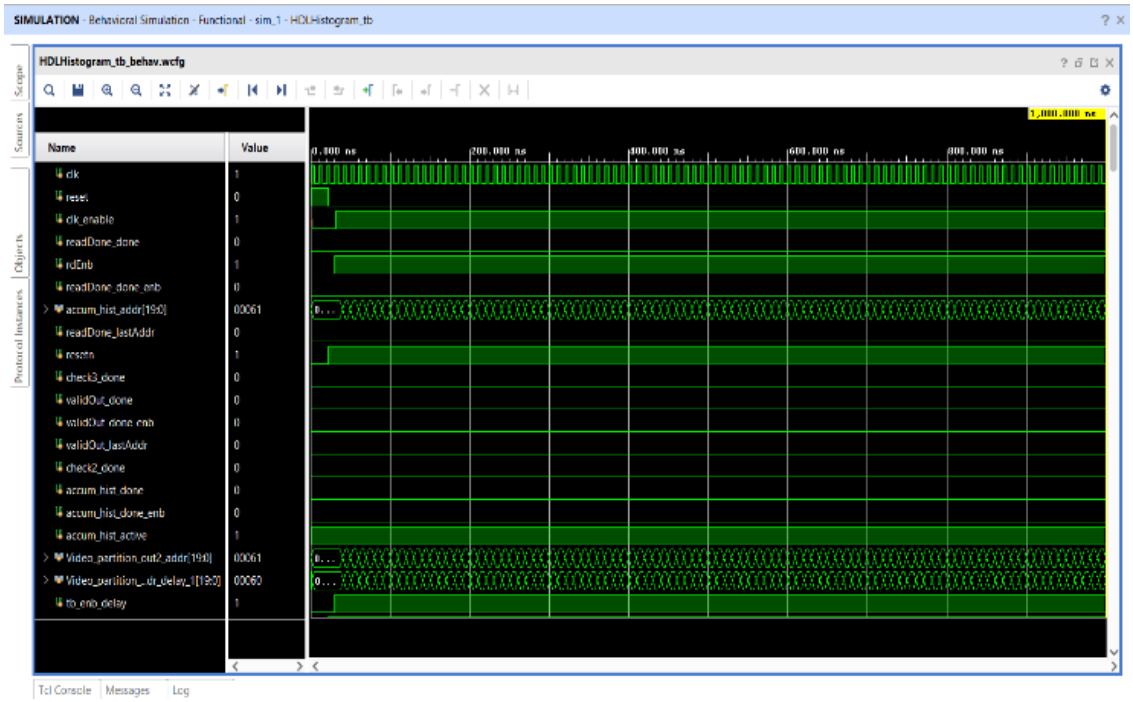


Figure 3.11. Simulation Results of HDL Histogram Equalization

Validation of the created MATLAB/Simulink model on hardware was planned as the next step at this point in the investigation. Cosimulation using the HDL Verifier tool was intended to carry out the verification for this purpose. Nevertheless, the cosimulation model generation procedure encountered an error due to the presence of nested record or structure types at the Design Under Test (DUT) interface, displaying the message 'Cosimulation model generation for nested record or structure types at the DUT interface is not supported.' The cosimulation process was unable to continue because the current version of HDL Verifier does not accept such data structures, and there is currently no workaround for this limitation. As a result, HDL Verifier-based cosimulation was not appropriate for this investigation, and the validation was restricted to simulation results from the Simulink environment and hardware description simulations carried out in the Xilinx Vivado environment.

3.2 Contrast Limited Adaptive Histogram Equalization (CLAHE)

A more sophisticated, confined form of conventional histogram equalization is called Contrast Limited Adaptive Histogram Equalization (CLAHE). Histogram equalization is applied independently within each of the smaller blocks (64x64 pixels) that make up the image in this method. This enables the independent enhancement of contrast in each local region. CLAHE works particularly well in photos that include shaded parts, uneven lighting, or were taken in low light. The capacity of CLAHE to restrict contrast enhancement by a predetermined "contrast limit," which avoids over-amplification of noise and artifacts, is one of its most crucial characteristics. This is essential for applications like satellite images, night vision, and medical imaging where visual noise must be reduced. Although the CLAHE method produces better images, its implementation on FPGA systems requires more computational power and hardware resources since it requires block-based processing and more memory. However, the advantages are substantial, yielding images with maintained fine details and improved local contrast. As a result, CLAHE is the recommended technique when both local and global contrast enhancements are required, particularly when maintaining the image's visual integrity and detail is crucial. In delicate applications like diagnostic imaging, it is frequently the default option[40].

3.2.1 Model Based Desing of CLAHE

A pre-processing method called adaptive histogram equalization (AHE) is used to enhance image contrast. Tile creation, histogram equalization, and bicubic interpolation are the three primary components of the CLAHE method. The input image is first separated into tiles,

which are parts of the image. For instance, four tiles are created from the input image. After that, each tile undergoes histogram equalization using a predetermined clip limit. In order to create an output image with improved contrast, the resultant tiles are then blended using bicubic interpolation.

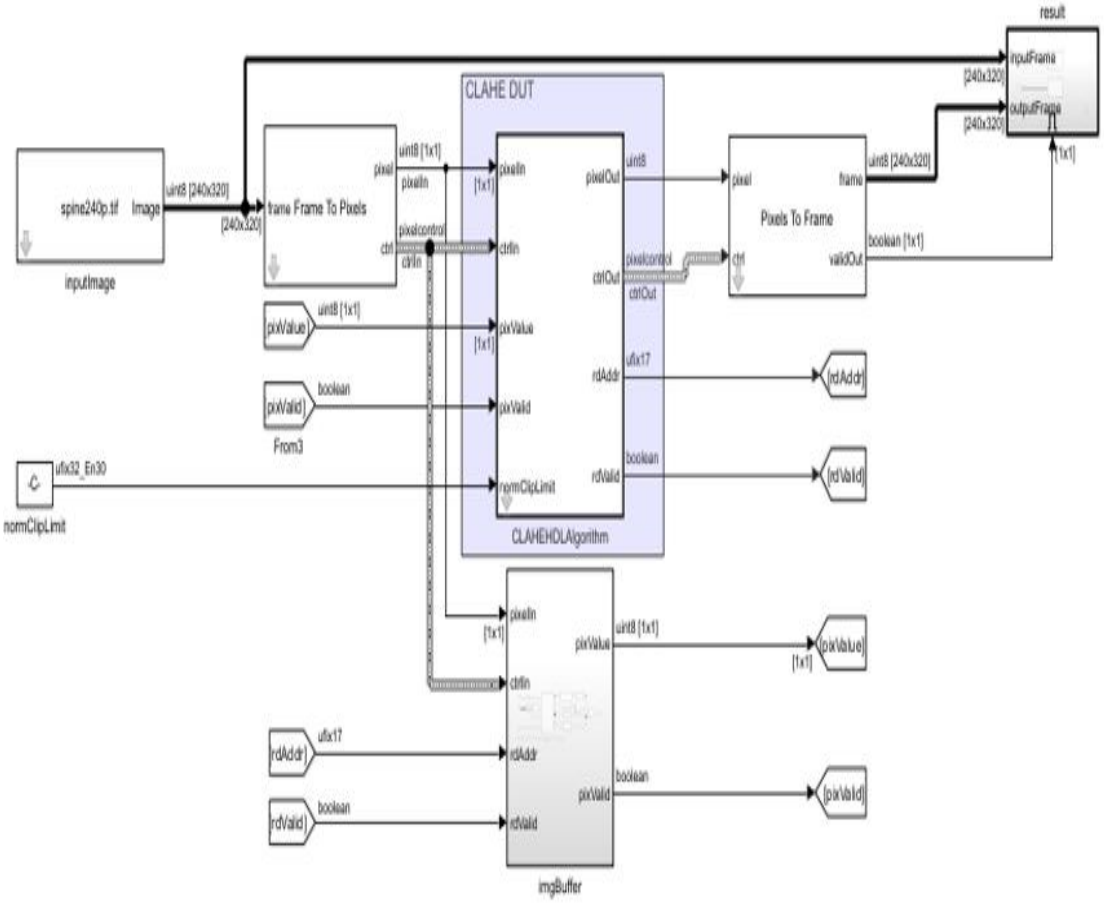


Figure 3.12. Contrast Limited Adaptive Histogram Equalization Block Diagram

A tile generation subsystem that divides the input image into horizontal and vertical tile typically eight in each direction is depicted in the figure. The pixel control bus is used to choose tiles, and the number of tiles is divided by the rows and columns of the image to determine their size. The ROI Selector block generates distinct pixel streams for every horizontal tile and processes vertical tiles in parallel. Every vertical tile is sequentially routed to a distinct processing pipeline. If the image does not divide equally, it is padded symmetrically. Tile sizes must be even integers.

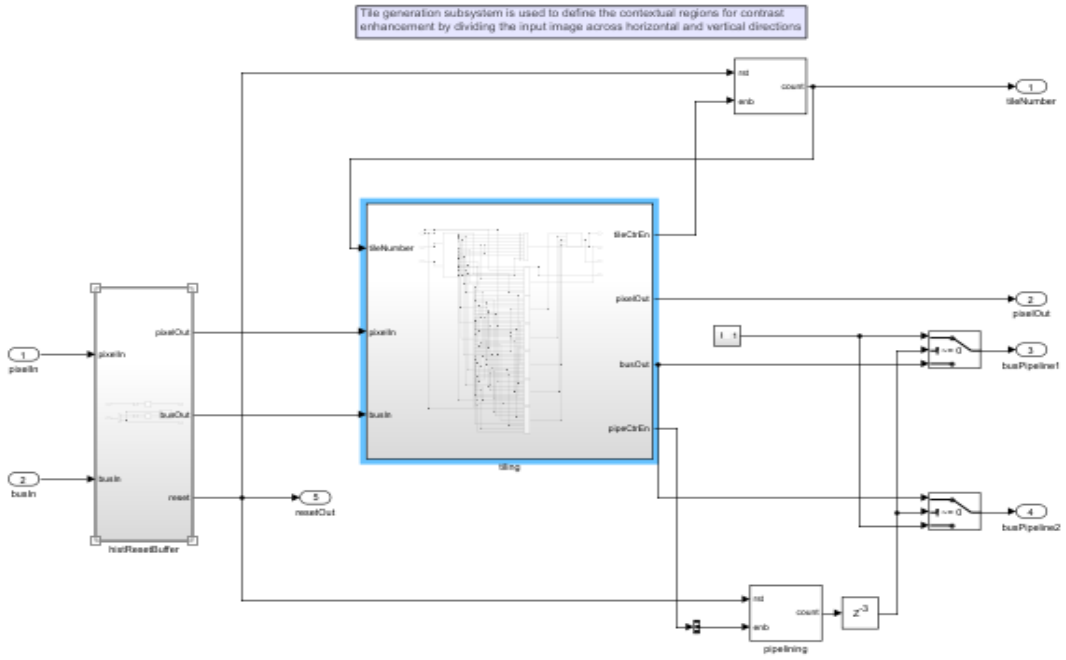
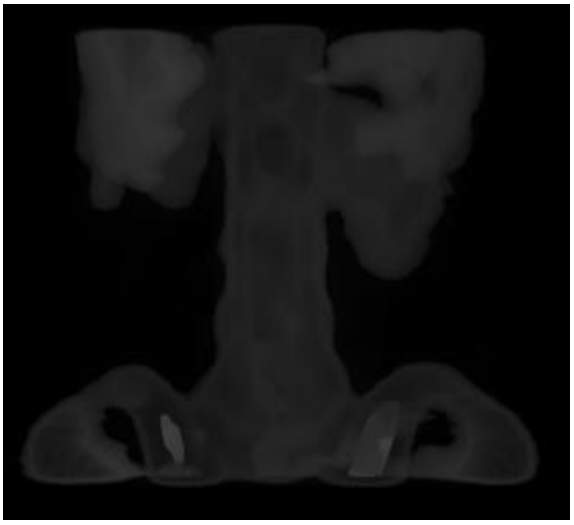


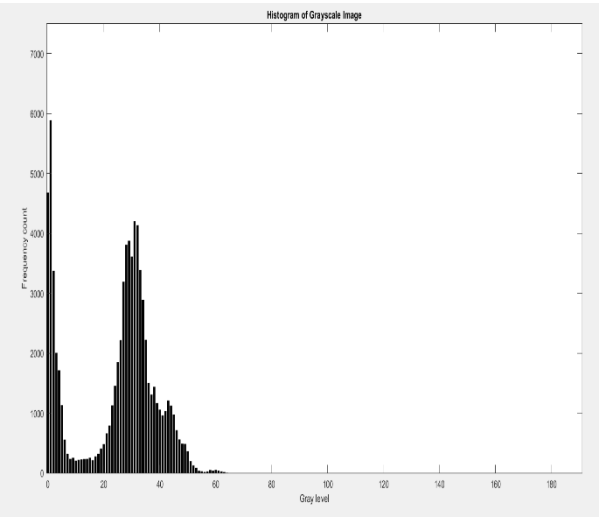
Figure 3.13**.** Generation Subsystem of CLAHE block diagram

3.2.2 Simulation in Simulink of CLAHE

The input Image block in this example specifies the file location for the 240 by 320 pixel input picture. An 8-bit input width, which corresponds to the uint8 data type, is used to represent the image pixels. The computed tile size is 30-by-40 pixels, or 1200 pixels per tile, with 8 tiles along each axis. 256 is the histogram bin count.



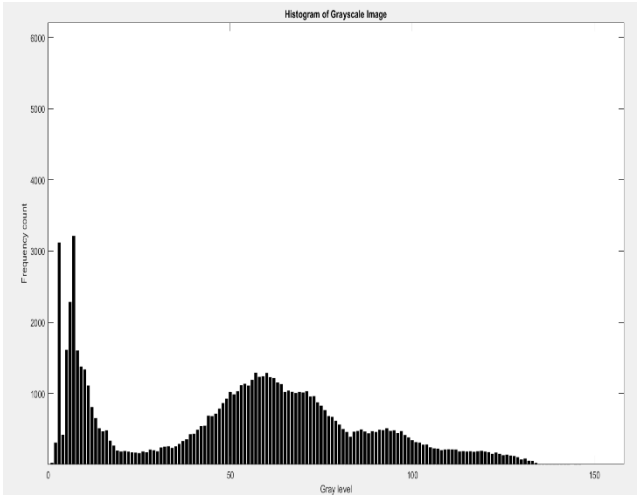
(a)



(b)



(c)



(d)

Figure 3.14. (a) Original Image [40] (b) Histogram Graph of Original Image (c) Result of Image CLAHE [40] (d) Histogram Graph of Result Image

The CLAHE model's input and output images are shown in this graphic. The output has improved contrast without becoming overly saturated. The findings of the CLAHE HDL model nearly match the adaphisteq function in MATLAB, with only minor pixel differences.

3.2.3 HDL Code Generation of CLAHE

In order to generate VHDL and SystemVerilog code from model-based designs in Simulink, the “HDL Coder” toolbox in Simulink is used. This toolbox includes two main components: “HDL Coder Advisor” and “HDL Workflow Advisor.” First, the design is validated using the “HDL Coder Advisor.” Once the design successfully passes all checks, the “HDL Workflow Advisor” is then used. The output of the “HDL Workflow Advisor,” which completes all processes, is shown in Figure 2.5.

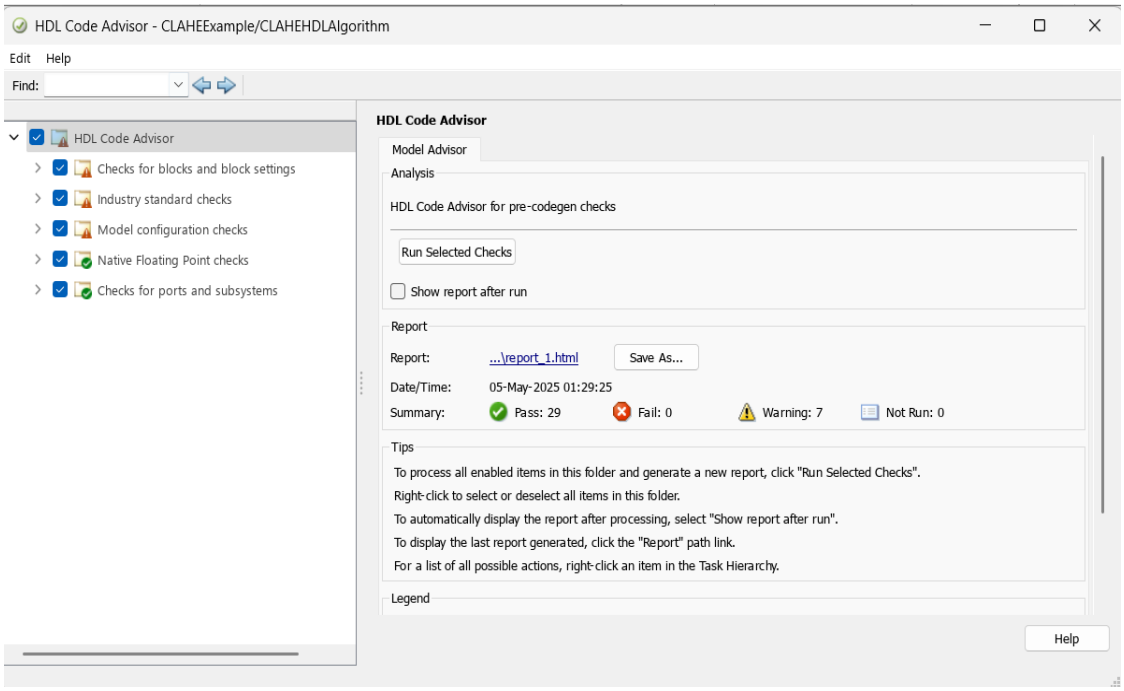


Figure 3.15. HDL Workflow Advisor Result

After testing that the blocks in the Simulink model are compatible with HDL Coder, the CLAHEHDLAlgorithm block is right-clicked to generate the HDL code. Then, the HDL code option is clicked, followed by the HDL Coder properties option.

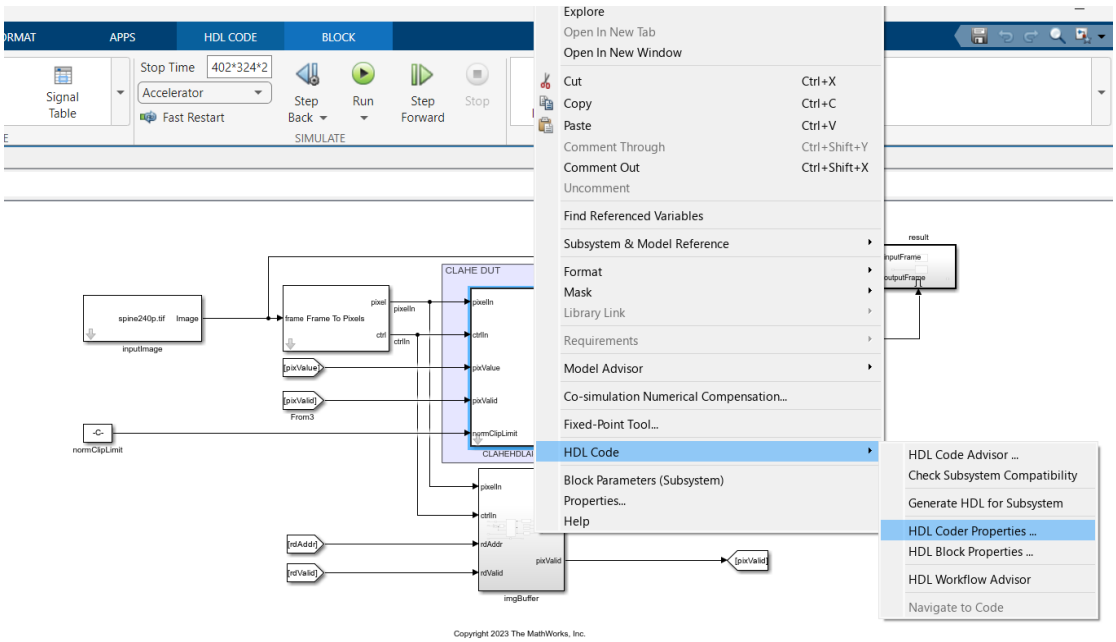


Figure 3.16. HDL coder Properties Selection

From the window that opens, SystemVerilog is selected as the HDL code language to be generated. Then, OK is clicked to generate the code. After these steps, the code is generated and prepared ready for Vivado.

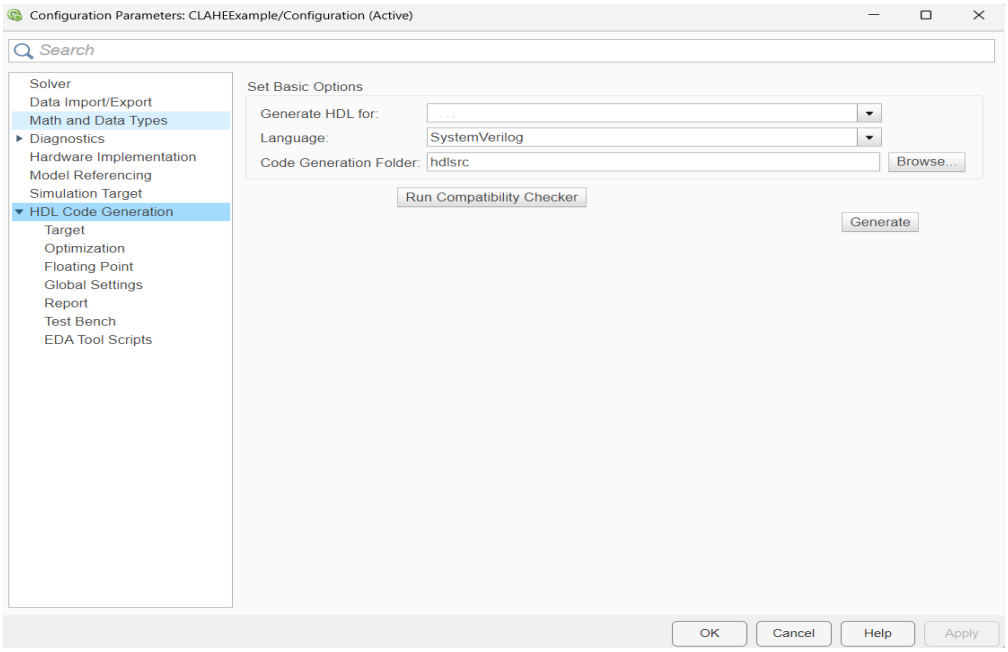


Figure 3.17. HDL coder Generation window

3.2.4 Synthesis and Simulation in Vivado of CLAHE

These source files were added to Xilinx Vivado 2024.2 following the acquisition of the SystemVerilog codes for Contrast Limited Adaptive Histogram Equalization. The RTL schematic of the programs was then produced after choosing the "Run Synthesis" option. There are numerous subsystems in the RTL schematic, but just the main circuit is shown because it is deemed unnecessary to display each one separately.

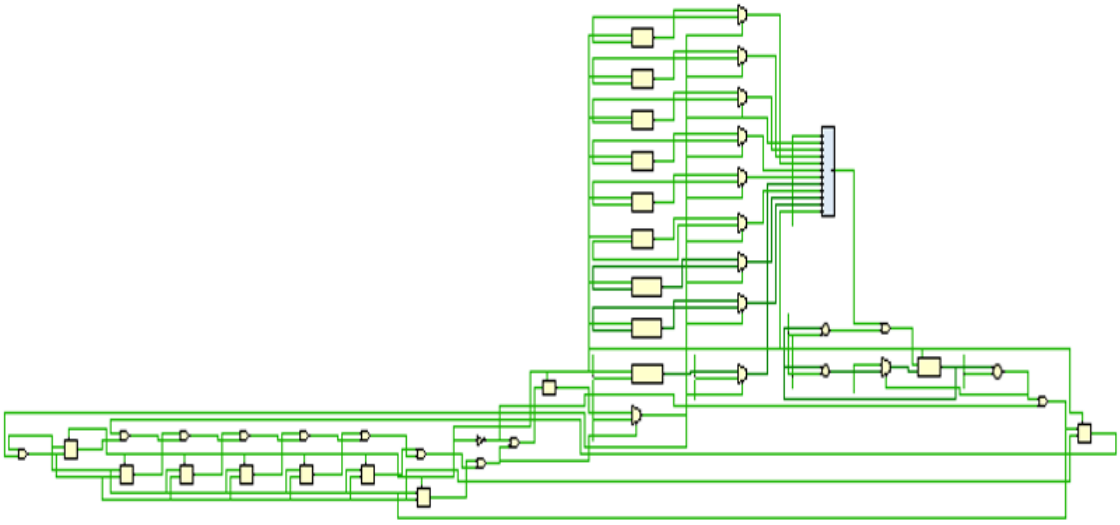


Figure 3.18. RTL Schematic of Contrast Limited Adaptive Histogram Equalization

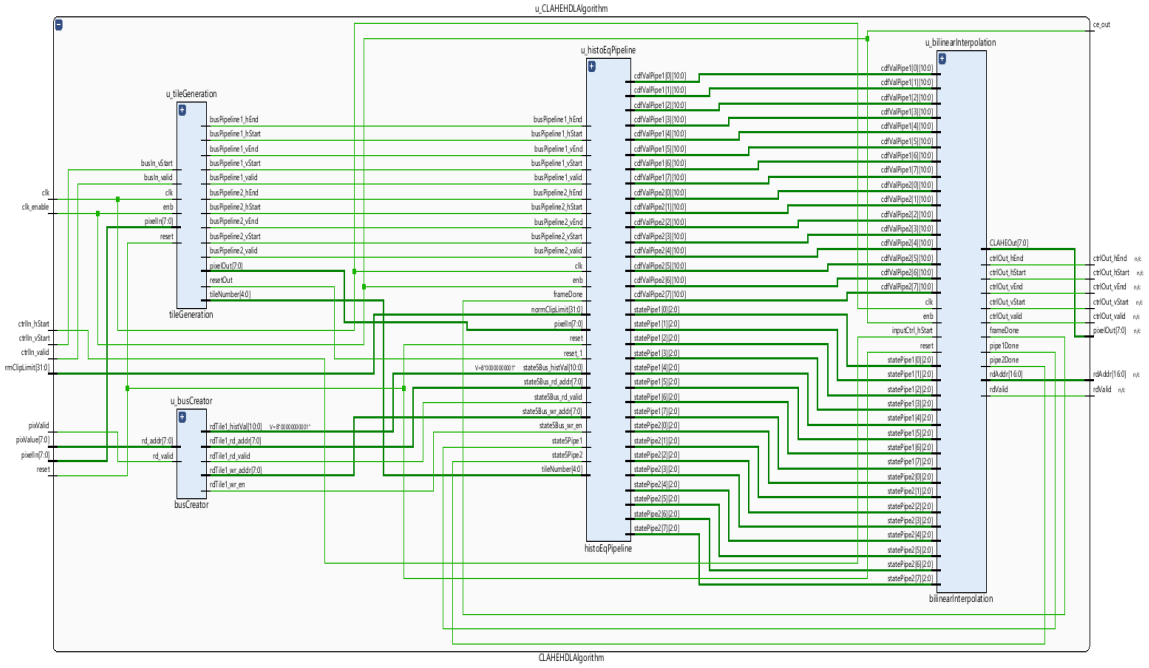


Figure 3.19. RTL Schematic Subsystem of Contrast Limited Adaptive Histogram Equalization

The testbench code for Contrast Limited Adaptive Histogram Equalization (CLAHE) was simulated in Xilinx Vivado, and the test results are shown in Figure 3.19.

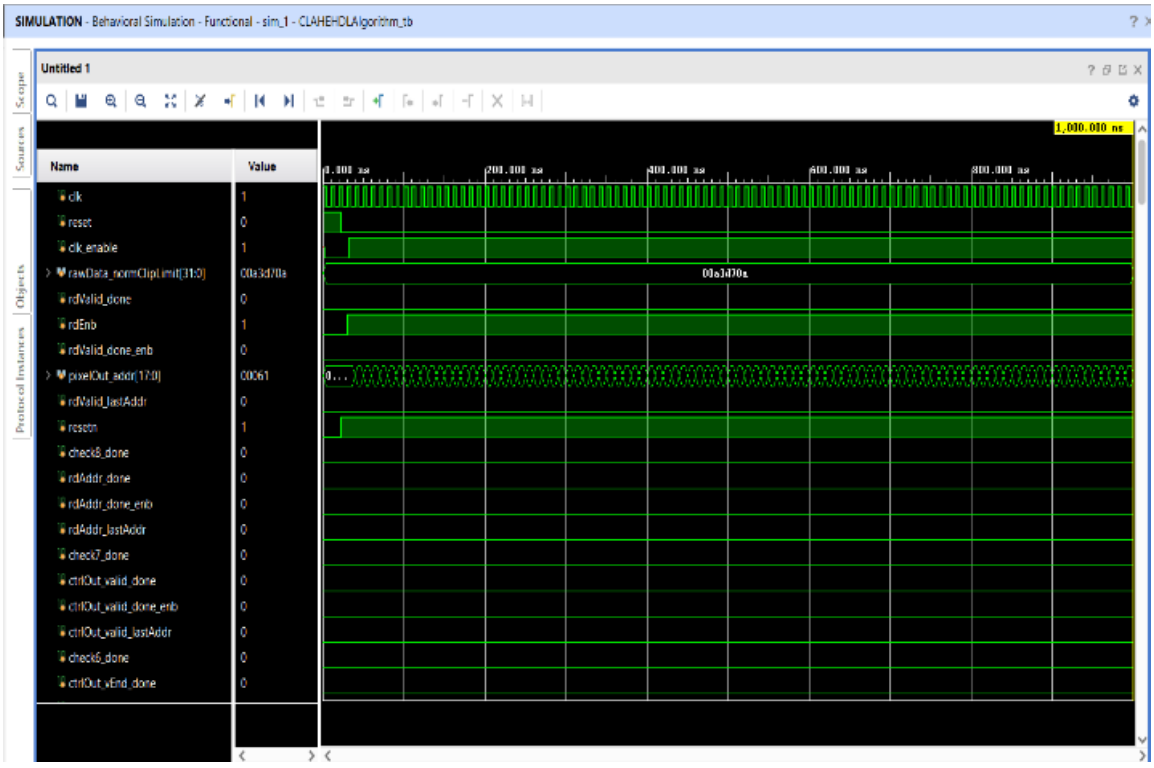


Figure 3.20. Simulation Results of Contrast Limited Adaptive Histogram Equalization

The CLAHE (Contrast Limited Adaptive Histogram Equalization) Simulink model, which was created as part of this thesis, was supposed to be cosimulated using HDL Verifier. The following problem occurred during this process: “Could not find ModelSim executable vsim. Make sure that ModelSim is installed on this machine and its executables are on the system path.” Either the ModelSim program is not installed on the computer or its executable files have not been properly added to the system path variable, as indicated by this error. The required installation and configuration could not be properly finished in spite of several efforts. This made it impossible to use HDL Verifier for ModelSim-based cosimulation. As a result, the simulation results acquired in the Simulink environment and the simulations conducted using Vivado were the only hardware-oriented verifications of this model.

4. COST ANALYSIS

MATLAB - Simulink, Xilinx Vivado, and other plugins are provided free of charge by Dokuz Eylul University. In addition, access to the articles in the references is also provided free of charge thanks to the Dokuz Eylul University subscription. The cost of the project is presented in the table below.

Components & Programs	Prices
MATLAB	Provided by University
Xilinx Vivado	Free

Table 4.1. Cost Analysis

4.1 Energy Consumption

This energy consumption, presented using MATLAB - Simulink, is compatible with the energy consumed by the lighting devices used to charge the computer used and to illuminate the work area.

4.2 Impacts on Healthcare

Histogram equalization is a frequently used method in image processing and can have a variety of impacts on healthcare. The importance of image enhancement in healthcare is increasing and forms the basis of accurate diagnosis and treatment processes. Histogram equalization can help make medical images clearer and more distinct, which can make diagnoses more accurate and treatment plans more effective. Improving images used in medical imaging techniques, especially x-ray, MRI and CT, can provide benefits in many areas, from pathology diagnoses to surgical planning. Histogram equalization can help surgeons and medical professionals make accurate diagnoses by making details in images appear better. However, caution should be exercised when using this technique and possible side effects should be considered. For example, histogram equalization may increase noise or lose certain details in images in some cases. Therefore, it is important that image enhancement techniques such as histogram equalization are carefully evaluated and used appropriately in healthcare.

4.3 Environmental Healthcare

This depends on factors such as energy consumption and material usage. This FPGA-based implementation can achieve lower power consumption than a traditional computer-based implementation because FPGAs can perform processing more efficiently by taking advantage of parallel processing capability and optimized hardware architectures. This generally means lower energy consumption and therefore less severe impact. Additionally, FPGAs generally have smaller forms, which can result in less material usage. However, other factors such as production and waste management should also be considered. As a result, it can be reduced by optimizing the design and energy efficiency.

4.4 Social Impacts

The social impact of this project could be quite large. This project can be used in many fields such as medical imaging, security systems, automotive industry and video processing. Particularly in medical imaging applications, obtaining clearer and more distinct images can help patients receive accurate diagnosis and treatment. In security systems, image enhancement can help detect events more accurately and identify dangerous situations in advance. In the automotive industry, it can be used to increase driving safety. This project can create a safer and healthier living environment, contributing to the general safety and health of the community.

4.5 Ethical Impact

In an ethical study, the necessary security measures must be taken to protect the data from unauthorized access and to use it only for specified purposes. It is also ethically important to distribute the developed technology in an accessible and fair manner, that is, to reach a wide range of users without discriminating in favor of a particular group or individual. In this context, developers must adopt a fair and transparent approach that considers social benefit.

4.6 Standards

Throughout the project, industrial standards for image processing and image file formats will be adhered to. The Joint Photographic Experts Group established JPEG as a digital image format. 1994 saw the standardization of this format as ISO 10918-1[41].

IEEE Std 1364-1995, which became the Verilog hardware description language (HDL) standard, was published in 1995. Following the completion of the standardization process, the IEEE P1364 Working Group began soliciting input from IEEE 1364 users globally in order to improve and adjust the standard as needed. This prompted a five-year endeavour to create IEEE Std 1364-2001, a substantially improved Verilog standard. In 2001, as Accellera started to standardize SystemVerilog, more problems were found that would have caused incompatibilities between Verilog 1364 and SystemVerilog. To help assure uniform resolution of such issues, the IEEE P1364 Working Group was formed as a subcommittee of the SystemVerilog P1800 Working Group. The outcome of this cooperative effort is IEEE Std 1364-2005, the standard [42].

5. WORK PLAN

Table 5.1 and Table 5.2, respectively, display the Gantt charts that show the project process and milestones for the Fall and Spring terms. The histogram equalization project's successive stages—literature review, Simulink model creation, HDL code generation, FPGA simulation, and system verification—are depicted in these figures. The project's methodical approach is reflected in the planned timeline, which also shows steady progress toward predetermined goals.

Fall Semester	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Literature Review of Digital Down Converter and Its Sub-blocks	■	■	■	■										
Literature Review of Model Based Design & FPGA					■	■	■	■						
Analyze and Simulation on MATLAB									■	■	■	■		
Writing Procedure													■	■

Table 5.1. Gantt Chart of Fall Terms Work-Flow

Present Semester	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Learning HDL Code Generation via Simulink	■	■	■	■										
Simulation Process					■	■								
Logic Synthesis and Circuit Implementation for FPGA							■	■	■	■	■	■		
Writing Procedure													■	■

Table 5.2. Gantt Chart of Spring Terms Work-Flow

6. CONCLUSION AND FUTURE WORKS

6.1 Conclusion

This thesis used MATLAB Simulink and HDL-compatible tools to design and implement an image improvement system based on histogram equalization techniques. In particular, two models from MathWorks' Vision HDL Toolbox—Historic Equalization and Contrast Adaptive Histogram Equalization (CLAHE)—were examined, modified, and simulated in the Simulink environment. These models' algorithmic nature and compliance with HDL code generating methods make them especially well-suited for hardware implementation.

In Simulink, the designs were simulated and functionally verified before being converted into SystemVerilog code using MATLAB HDL Coder. In order to assess the

hardware description code's structural viability and usefulness, it was then synthesized and simulated using the Xilinx Vivado Design Suite. The accuracy and effectiveness of the hardware architecture that was produced were validated using RTL schematic views and simulation results.

The final design demonstrates a hardware-ready solution for real-time image enhancement applications. Given the successful synthesis and simulation of the system, the proposed architecture is now fully prepared for implementation on FPGA hardware, particularly targeting platforms such as the Xilinx Spartan-7 series. The use of FPGA offers significant advantages, including high-speed processing, low power consumption, and suitability for embedded vision systems.

6.2 Future Works

The physical implementation of the CLAHE and histogram equalization algorithms, which were created and ready for FPGA adaption in this thesis, on an FPGA platform like the Xilinx Spartan-7 is one of the important future directions. This would make it possible to test in real time using live picture or video inputs, which would make it appropriate for assessing performance metrics like processor speed, latency, and hardware resource usage in real-world scenarios. Furthermore, it may be advantageous to maximize hardware resource efficiency and minimize power consumption by utilizing strategies like pipelining, parallel processing, and fixed-point arithmetic refinement.

For portable and embedded image processing applications, the system could be improved in efficiency. Lastly, to improve the system's resilience and suitability for practical uses, it would be acceptable to incorporate the image enhancement module with other image processing pipelines and include the capacity to dynamically adjust to changing lighting circumstances.

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